
Local Policies for Graph-Structured Markov Decision Processes

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Abstract

We study a cooperative form of multi-agent reinforcement learning with state space dynamics and agent interaction controlled by an underlying graph. Each agent has a local state and action, the evolution of the local state depends only on the states and actions in the 1-hop neighborhood defined by the graph. Structured dynamics of this type arise in various applications, including network resource allocation, co-operative games, epidemic control, and wireless scheduling. The global state-action space scales exponentially in the number of agents, so that computing global optimal policies is intractable in the worst-case. We study conditions under which it is possible to approximate the optimal policies by a local policy for each agent that depends only on states associated with nodes within its m -hop neighborhood. By controlling the propagation of influences via a Dobrushin-type stability matrix, we establish that globally optimal policies can be approximated by local policies with sub-optimality gap decaying exponentially in m .

1. Introduction

Markov decision processes (MDPs) provide a formalization of sequential decision-making under uncertainty, and have proven useful in a wide variety of applications. Optimal policies are characterized by the Bellman principle, and in theory, can be computed either via exact dynamic programming when the model is known (e.g. (Puterman, 2014; Bertsekas, 2025)), or approximated using samples via reinforcement learning (RL) algorithms (Bertsekas, 2025; Sutton & Barto, 2018). In either setting, the curse-of-dimensionality manifests itself via growth of the state and/or action spaces. Accordingly, there is a large body of work in approximate

dynamic programming and/or RL that focuses on particular structured classes of MDPs, and associated schemes for computing near-optimal policies.

In many application domains, there exists additional structure in the form of an underlying graph. This is true, for instance, in controlling epidemics and wildfires (Peyrard et al., 2007; Haksar & Schwager, 2018); wireless scheduling (Feriani & Hossain, 2021); and controlling robots and autonomous vehicles (Kober et al., 2013; Shalev-Shwartz et al., 2016). In this paper, we study MDPs with state and action spaces that decompose according to an underlying graph, and for which the state dynamics respect the graph structure. Associated with each node is an agent, along with a local state, a local action, and a local reward function that depends only on the local state-action pair. Moreover, any agent’s actions have a direct effect only on its own states, and those of its immediate neighbors, and the state dynamics depend only on the local neighborhood of state-action pairs. We study the cooperative setting, in which agents share the common goal of maximizing the discounted long-term reward summed across all of the agents.

One concrete instantiation is the problem of epidemic control (Peyrard et al., 2007; Haksar & Schwager, 2018). In this setting, each agent corresponds to a person, and the local state could be a binary variable indicating whether or not they are infected. An agent can take a range of possible actions, which (depending on the disease) could include taking medicine, vaccination, or self-isolation. Edges in the graph correspond to dependencies between individuals’ infection statuses; the probability of transition of a particular individual’s infection status directly depends on their neighborhood’s infection status and the control action taken. See Section 5 for further discussion of this example.

Now consider the problem of finding an optimal policy. A government agency might seek to maximize the long-term health of all the individuals, but it is infeasible to have global co-ordination among all individuals, along with disclosure of their private health information. Consequently, it is natural to seek policies that are m -hop local, meaning that every individual can make decisions based on information that depends only on individuals at distance at most m in the underlying graph.

For a large-scale networked system, it becomes intractable

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for any single agent, with limited memory and/or computational power, to implement a globally optimal policy. Thus, one is led to the following natural question:

Under what conditions can the globally optimal policy of an agent be well-approximated by a local policy, depending only on a local neighborhood of states and actions?

The main contribution of this paper is to study this question—and to provide some precise answers—for this broad class of networked multi-agent systems.

Our contributions. The main contribution of this paper is to give sufficient conditions for the existence of m -hop local policies that are good approximations to the globally optimal policy. In particular, our results show that the existence of near-optimal local policies depends on the *propagation of influences* in the underlying network MDP. In this way, we establish connections between approximation schemes for networked MDPs, correlation decay in graphical models (Gamarnik et al., 2014), and mixing time analysis of Markov chains (Levin & Peres, 2017).

Obtaining our theoretical guarantees involves tackling two broad challenges, both of which play an important role in the multi-agent RL literature. The first is that of *partial observability*: if agents do not use the entire state-space, how much do they lose? The second is that of *coordination* between the agents: how can they communicate with their neighborhood to collaboratively compute and implement an approximately optimal policy?

1. To focus on the effect of partial observability, we first study a simpler setting, involving a single active agent, where no coordination is required. Here our main result (Theorem 1) gives conditions under which there exists a m -local policy with sub-optimality gap that decays exponentially with m . Moreover, we describe how to compute this local policy in a space- and time-efficient manner.
2. Using insights from controlling partial observability errors in the single-agent case, we then turn to the multi-agent problem with k active agents. In Sections 4.3 and 4.4, we show the existence of an approximately-optimal policy that only requires each active agent to communicate and coordinate with their m -hop neighborhood, with a suboptimality gap that decays exponentially in m/k . The structure of the policy shows how agents can coordinate their actions to achieve approximately-optimal performance.

In both of our results, we provide explicit bounds on the sub-optimality gap relative to the Q -function of the globally optimal policy. This type of guarantee differs from past work that either restricts to the class of 0-hop local policies (Qu et al., 2020), or compares to an entropy-regularized

policy (Zhang et al., 2023). To the best of our knowledge, our results are the first to establish the existence of near-optimal local policies, as measured relative to the global optimum. In certain versions of the multi-agent set-up, our policies require agents to coordinate their actions with their neighbors. Such local co-ordination policies have not been studied before in the literature, and suggest multiple open directions for future work.

2. Related Work

In broad terms, the field of multi-agent RL focuses on strategic agents that interact with each other in dynamic (usually Markovian) environments. In the case of static environments, there is a large body of work on distributed cooperative optimization (Nedic & Ozdaglar, 2009; Olshevsky & Tsitsiklis, 2009) forming a solid foundation upon which several results for dynamic environments have been shown. There is also work on distributed forms of classic RL algorithms, including TD learning (Macua et al., 2014; Mathkar & Borkar, 2016).

Scalability issues for large-scale systems have been the focus of considerable work in the multi-agent RL literature (e.g., Littman, 1994; Busoniu et al., 2008)). There are various structured sub-classes of MDPs that allow for side-stepping the computational intractability induced by the exponentially large state-action space. One broad class are factored MDPs (Kearns & Koller, 1999; Guestrin et al., 2003), in which the state space and transitions are factored, whereas the action remains a global quantity. Osband & Van Roy (2014) studies optimal policy learning for a factored MDP; their method is efficient, but requires that all agents coordinate with every other agent. There is also work on weakly coupled MDPs (Meuleau et al., 1998), in which an ensemble of MDPs with independent transitions are coupled by resource constraints.

There is also a rich literature on partially observable systems and their tractability. Optimal policies for single-agent POMDPs are typically history-dependent, computationally intractable to compute in the worst case (Papadimitriou & Tsitsiklis, 1987), and require storage scaling exponentially with the horizon. There is a large literature on various practical schemes for approximating or computing optimal policies (Smith & Simmons, 2004; Sunberg & Kochenderfer, 2018; Silver & Veness, 2010; Shani et al., 2013; Golowich et al., 2023), as well as work studying decentralized planning and learning problems for POMDPs (Oliehoek et al., 2016; Katt et al., 2018).

Early work on graph-structured or networked MDPs include the papers (Peyrard & Sabbadin, 2006; Chorney et al., 2005; Peyrard et al., 2007). In the LQR version of this problem, Shin et al. (2023) prove the existence of a near-optimal

decentralized controller. Most related to our work are some recent papers (Qu et al., 2020; Zhang et al., 2023) on local policies for graph-structured MDP models. In particular, Qu et al. (2020; 2022) propose a decentralized method for optimizing over the class of 0-hop policies, meaning that the policy depends only on the agent’s local state, but do not provide guarantees on the suboptimality gap. On the other hand, Zhang et al. (2023) studies policies that are approximately optimal, but where the comparison is made to the fixed point of an entropy-regularized Bellman operator. When the regularization parameter is large, this fixed point is quite different from the standard optimal Q -function to which we compare.

3. Setup

Let G be a graph with n nodes, and consider a graph-structured MDP with the global state space $\mathcal{S}$, global action space $\mathcal{A}$ and reward function $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. Using the shorthand $[n] := \{1, \dots, n\}$, we assume that the state and action space are factored, i.e., $\mathcal{S} = \otimes_{i=1}^n \mathcal{S}_i$ and $\mathcal{A} = \otimes_{i=1}^n \mathcal{A}_i$ for all $i \in [n]$, where $\mathcal{S}_i$ is the local state space and $\mathcal{A}_i$ is the local action space. In addition, we assume that the reward function is additively separable, meaning that

$$R(s, a) = \sum_{i=1}^n R_i(s_i, a_i), \quad (1a)$$

where $R_i : \mathcal{S}_i \times \mathcal{A}_i \rightarrow \mathbb{R}$ is the local reward function of node i , depending only on the local state-action pair (s_i, a_i) .

Let $d_G(i, j)$ denote the shortest distance on the graph between two nodes i and j . For any integer $k = 0, 1, \dots, n-1$, we define the k -hop neighborhood

$$\mathbb{N}_i^k := \{j \in [n] \mid d_G(j, i) \leq k\}. \quad (1b)$$

Since node i lies at distance 0, it belongs to the set $\mathbb{N}_i^k$. With this notation, we assume that the transition kernel factorizes as

$$\mathbb{P}(s' \mid s, \mathbf{a}) = \prod_{i=1}^n \mathbb{P}_i(s'_i \mid s_{\mathbb{N}_i^1}, a_{\mathbb{N}_i^1}), \quad (1c)$$

i.e., the transition probabilities of each local state only depends on the local states and actions of its immediate neighborhood.

A policy π is a mapping from the global state space to the global action space of all the nodes, i.e., π is a $\mathcal{S} \rightarrow \mathcal{A}$ mapping. For a given policy π , the γ -discounted Q -function is given by

$$Q^\pi(s, a) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k R(s(k), a(k)) \mid (s(0), a(0)) = (s, a) \right],$$

where the global states and actions evolve in the following way: at any time k when the whole system is at the global state $s(k)$, the agents simultaneously choose the global action $a(k+1) \sim \pi(\cdot \mid s(k))$, and the global state evolves according to the transition kernel $s(k+1) \sim \mathbb{P}(\cdot \mid s(k), \mathbf{a}(k))$. A policy π^* is *globally optimal* if it maximizes its Q -function uniformly across all state-action pairs—that is, if

$$Q^{\pi^*}(s, a) \geq Q^\pi(s, a) \quad \text{for all } (s, a) \text{ and } \pi. \quad (2)$$

Define the Bellman optimality operator $\mathcal{B}^* : \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$

$$\mathcal{B}^*(Q)(s, a) = R(s, a) + \gamma \sum_{\tilde{s} \in \mathcal{S}} \mathbb{P}(\tilde{s} \mid s, a) \max_{\tilde{a} \in \mathcal{A}} Q(\tilde{s}, \tilde{a}). \quad (3)$$

Standard results (Puterman, 2014) guarantee the existence of such a globally optimal policy. Since the operator $\mathcal{B}^*$ is a γ -contraction with respect to the ℓ_∞ -norm, it has a unique fixed point Q^* , and this fixed point corresponds to the Q -function of any globally optimal policy. The globally optimal policy generally requires coordination between all the agents and full observability of the entire state space to implement. Moreover, the space and sample complexity of computing such a globally optimal policy scales with the size of the entire state-action space, rendering it impractical to implement.

4. Main Results

We now turn to the statements of our main results. To first build intuition, we begin in Section 4.1 by analyzing the case of a single agent. Here our main result (Theorem 1) shows how the approximate optimality of an m -hop policy can be controlled by a pairwise interaction matrix $\mathbf{C}$, (cf. equation (4a)); in particular, the spectral radius $\rho(\mathbf{C})$ controls the rate of decay. Building from this analysis, Section 4.3 addresses the simplest multi-agent setting of two active agents, and our main result (Theorem 2) gives a performance guarantee for m -hop co-operative policies. In Section 4.4, we generalize this result for k active agents. For the sake of brevity, we present a slightly weaker form of our error bounds; see the supplement for tighter results. Finally, in Section 4.2, we discuss the sole assumption required to prove our results and what it means for the underlying graph-structured MDP.

4.1. The case of the single active agent

In the simplest case, we have a structured MDP with only a single active agent. Without loss of generality (re-indexing as needed), we assume that agent 1 is active, with a non-trivial reward and action space, and that $R_j = \mathbf{0}$ and $|\mathcal{A}_j| = 1$ for all agents $j \in [n] \setminus \{1\} \equiv \{2, \dots, n\}$. For a given

integer $m \in [n]$, we are interested in finding an m -hop local policy for agent 1, meaning a policy that depends only on states in the m -hop neighborhood around node 1.

A central object in our analysis is the *pairwise influence matrix* $\mathbf{C} \in \mathbb{R}_+^{n \times n}$ with $(i, j)^{th}$ entry given by

$$\mathbf{C}_{i,j} := \max_{\substack{(s,a)_j, s-j, \\ (\tilde{s}, \tilde{a})_j, a-j}} d_{\text{TV}}(\mathbb{P}_i(\cdot | s, a), \mathbb{P}_i(\cdot | (\tilde{s}, \tilde{a})_j, (s, a)_{-j})), \quad (4a)$$

where d_{TV} denotes the total variation (TV) distance between two distributions.¹ The matrix entry $\mathbf{C}_{i,j}$ measures the influence of changing the local state and action of agent j on the transition kernel of node i . By definition of the TV distance, we always have the inclusion $\mathbf{C}_{i,j} \in [0, 1]$, and moreover, if agents i, j are not neighbors on the graph G , then $\mathbf{C}_{i,j} = 0$. Therefore, we can understand $\mathbf{C}$ as a type of weighted adjacency matrix. In addition, our result depends on the k -hop shell around node i , given by

$$\mathbb{T}_i^k := \{j \in [n] \mid d_G(j, i) = k\}. \quad (4b)$$

Theorem 1. *Consider a γ -discounted graph-structured MDP with pairwise influence matrix $\mathbf{C}$ satisfying the spectral radius bound $\gamma\rho(\mathbf{C}) < 1$. Then there exists an m -hop policy $\tilde{\pi}^{(m)} : \mathcal{S}_{\mathbb{T}_1^m} \rightarrow \mathcal{A}_1$ with sup-norm error $\|Q\tilde{\pi}^{(m)} - Q^*\|_\infty$ at most*

$$\frac{2\text{osc}(R_1)\|\mathbf{C}\|_\infty}{1-\gamma} \sum_{\ell \in \mathbb{T}_1^m} ((\gamma\mathbf{C})^m (\mathbf{I} - \gamma\mathbf{C})^{-1})_{(1,\ell)}, \quad (5)$$

where $\text{osc}(R_1) := \max_{s_1, a_1} R_1(s_1, a_1) - \min_{s_1, a_1} R_1(s_1, a_1)$.

See Section A for the proof.

Theorem 1 establishes the existence of an approximately optimal policy that depends only on the m -hop neighborhood of agent 1. Our analysis also shows that this policy can be computed in an efficient way. In particular:

- Given knowledge of the state transition kernels for each state in $\mathcal{S}_{\mathbb{T}_1^m}$, the policy $\tilde{\pi}^{(m)}$ can be computed in polynomial time with the space complexity $|\mathcal{S}_{\mathbb{T}_1^m}| |\mathcal{A}_{\mathbb{T}_1^m}|$.
- When state transitions are not known, the policy $\tilde{\pi}^{(m)}$ can be approximated from samples using a fitted Q -iteration (FQI) over the m -neighborhood.

The key component of the performance bound (5) is the term $((\gamma\mathbf{C})^m (\mathbf{I} - \gamma\mathbf{C})^{-1})_{(1,\ell)}$: it characterizes the long-range effect of ignoring the local states for each node ℓ

¹For use in the sequel, we have defined the matrix $\mathbf{C}$ for the general multi-agent setting; for the single agent setting under consideration in this section, only agent 1 has a non-trivial action space.

outside the m -ball surrounding node 1. Given our bound on the spectral radius, the term $(\gamma\mathbf{C})^m$ decays exponentially in m , and reflects the fact that any state changes outside the m -neighborhood require m steps to impact the reward of agent 1.

As shown in the proof, **Theorem 1** can be strengthened by introducing the *m -hop influence matrix* with entries

$$\mathbf{H}_{i,j}^{(m)} := \max_{\substack{s, \tilde{s} \\ a, \tilde{a}}} d_{\text{TV}}\left(\mathbb{P}_j(\cdot | s, a), \mathbb{P}_j(\cdot | (s, a)_{\mathbb{T}_i^m}, (\tilde{s}, \tilde{a})_{(\mathbb{T}_i^m)^c})\right). \quad (6a)$$

This quantity tracks the effect of changing the local states and actions outside the m -hop neighborhood of agent i on the state of node j . Note that $\mathbf{H}_{i,j}^{(m)} = 0$ for any j such that $d_G(i, j) < m$. With this definition, we show in the proof that the performance gap $\|Q\tilde{\pi}^{(m)} - Q^*\|_\infty$ is at most

$$\underbrace{\frac{2}{1-\gamma} \text{osc}(R_1) \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} ((\gamma\mathbf{C})^m (\mathbf{I} - \gamma\mathbf{C})^{-1})_{(1,\ell)}}_{\equiv U^{(m)}} \quad (6b)$$

which is a refinement of the bound (5).

Proof sketch: So as to provide intuition, let us describe some key ideas from the proof; see Section A for the technical argument itself. **Theorem 1** guarantees the existence of an m -hop policy $\tilde{\pi}^{(m)}$ with the prescribed optimality guarantees; the proof itself is explicit and constructive in nature. For any hopsize $m = 0, 1, \dots$, we define the projection operator

$$(\Pi^{(m)}Q)(s, a_1) := Q(s_{\mathbb{T}_1^m}, \bar{s}_{n \setminus \mathbb{T}_1^m}, a_1), \quad (7)$$

which fixes all states outside the m -hop neighborhood to some fixed state $\bar{s}$. By construction, the operator $\Pi^{(m)}$ is non-expansive with respect to the ℓ_∞ -norm. As a result, the projected Bellman operator $\Pi^{(m)} \circ \mathcal{B}^*$ is γ -contractive in the ℓ_∞ -norm. By the contraction mapping principle, there is a unique $Q^\dagger$ satisfying the fixed point relation $\Pi^{(m)}(\mathcal{B}^*(Q^\dagger)) = Q^\dagger$. Our analysis applies to any policy $\tilde{\pi}^{(m)}$ that is greedy with respect to $Q^\dagger$; since the projected fixed point $Q^\dagger$ depends only on variables on the m -hop neighborhood, any such greedy policy is m -hop local by definition. Given knowledge of the reward and transition dynamics, the function $Q^\dagger$ can be computed by the standard fixed point iteration $Q_{t+1}^{(m)} = \Pi^{(m)}(\mathcal{B}^*(Q_t^{(m)}))$. By the γ -contractivity, the iterates $Q_t^{(m)}$ converge geometrically fast to $Q^\dagger$.

The bulk of our technical effort is devoting to upper bounding the Bellman residual sup-norm error $\|\mathcal{B}^*(Q^\dagger) - Q^\dagger\|_\infty$

of the projected fixed point, and we do so by first bounding the difference

$$\Delta_{t+1}^{(m)}(s, a_1) := \mathcal{B}^* Q_t^{(m)}(s, a_1) - Q_{t+1}^{(m)}(s, a_1)$$

along the iterates of the algorithm. The key technical step establishes the uniform upper bound

$$\Delta_{t+1}^{(m)}(s, a_1) \leq \text{osc}(R_1) \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \left[\gamma^m \mathbf{C}^m (\mathbf{I} - \gamma \mathbf{C})^{-1} \right]_{(1,\ell)},$$

Observe that the right-hand side of this bound is independent of the iteration number t , so that we can take the limit as $t \rightarrow \infty$. Doing so leads to a bound on the Bellman residual of the projected fixed point $Q^\dagger$, as in equation (6b).

4.2. Spectral Radius and Graph Topology

Let us discuss the spectral radius bound $\gamma \rho(\mathbf{C}) < 1$ under which Theorem 1 is applicable. From its definition (4a), the pairwise influence matrix $\mathbf{C}$ is graph-structured, with nonzeros only for node pairs (i, j) that are connected in the graph; for any such edge, the associated weight $\mathbf{C}_{i,j}$ also depends on the transition structure. In the sequel (Section 5), we give numerical results that illustrate how both the graph topology and interaction strengths affect the decay rates.

The spectral radius can be upper bounded by any matrix norm, so that (for instance) it satisfies the upper bounds

$$\rho(\mathbf{C}) \leq \min \left\{ \underbrace{\max_{j \in [n]} \sum_{i=1}^n |\mathbf{C}_{i,j}|}_{\equiv \|\mathbf{C}\|_1}, \underbrace{\max_{i \in [n]} \sum_{j=1}^n |\mathbf{C}_{i,j}|}_{\equiv \|\mathbf{C}\|_\infty} \right\}. \quad (8a)$$

Since $\mathbf{C}$ is not symmetric in general, both upper bounds are useful. More crudely, the inequality (8a) implies the (very conservative) upper bound $\rho(\mathbf{C}) \leq d_{\max} \|\mathbf{C}\|_{\max}$, where $d_{\max}$ is the maximum degree of the graph and $\|\mathbf{C}\|_{\max}$ is the maximum entry of $\mathbf{C}$. (Recall that by definition of the TV norm, we always have the upper bound $\|\mathbf{C}\|_{\max} \leq 1$.) Thus, the spectral radius can be controlled by the maximum degree of the graph G and the maximum interaction strength between any two agents.

We can obtain a more refined dependence by exploiting bounds from spectral graph theory. Let $\mathbf{A}$ be the adjacency matrix of the graph G , we always have the entrywise upper bound

$$\mathbf{C} \preceq_{\text{e.w.}} \|\mathbf{C}\|_{\max} \mathbf{A}. \quad (8b)$$

Since both matrices have non-negative entries, the Perron-Frobenius theorem implies that $\rho(\mathbf{C}) \leq \|\mathbf{C}\|_{\max} \rho(\mathbf{A})$. Therefore, the structure of the underlying graph directly affects $\rho(\mathbf{C})$, which controls the rate of exponential decay in all of our suboptimality gaps. The effect of the spectral

radius of the adjacency matrix has been noted before in various network problems (e.g., (Van Mieghem et al., 2008; Duchi et al., 2012)).

As noted previously, we always have the worst-case bound $\rho(\mathbf{C}) \leq d_{\max}$ where $d_{\max}$ is the maximum degree, and this bound is achieved with equality by a $d_{\max}$ -regular graph (meaning that each node has exactly degree $d_{\max}$). For graphs without cycles (known as trees; see panel (a) in Figure 1 for an example), the spectral radius admits the sharper upper bound $\rho(\mathbf{C}) \leq 2\sqrt{d_{\max} - 1}$. Thus, our theory suggests that tree graphs have a faster decay in m than $d_{\max}$ -regular graphs, and this effect is present in our simulations (compare the tree in column (a) to the grid in column (c) in Figure 1).

4.3. The multi-agent problem

In Section 4.1, we established the existence of approximately-optimal policies that only use information from an m -hop neighborhood in the case of a single active agent. Here we extend these results to the case where k agents have nonzero rewards and non-trivial actions. For the sake of exposition, we first discuss the two-agent case before generalizing it to the k -agent case. Without loss of generality in the two-agent case, we assume that only agents 1 and 2 have nonzero rewards and non-trivial actions. The broad conclusion remains the same: for a given global state configuration, each agent can compute their optimal action in terms of Q -functions that depend only on their m -hop neighborhood, but with the subtlety that some form of coordination can be required in choosing optimal actions.

We define an m -hop cooperative policy $\tilde{\pi}^{(m)} : \mathcal{S} \rightarrow \mathcal{A}$ to be a policy that is greedy with respect to the sum of Q -functions, each of which only depends on an m -hop neighborhood. In analytical terms, we write

$$\tilde{\pi}^{(m)}(s) \in \arg \max_{a'} \sum_{i=1}^2 \tilde{Q}_i((s, a')_{\mathbb{N}_i^m}), \quad (9)$$

for some set of Q -functions $\{Q_i\}_{i \in [n]}$, where each Q_i is a function only of agent i 's m -hop neighborhood.

While the action of any agent j does depend on the global state s , there is additional local structure here: each agent only needs to communicate with its $2m$ -hop neighborhood to find what its own action to be played is. To see this fact more clearly, let $\tilde{Q}(s, a) := \sum_i \tilde{Q}_i((s, a)_{\mathbb{N}_i^m})$ for brevity. Then the optimality gap of action a_j compared to a'_j equals

$$\begin{aligned} \tilde{Q}(s, a) - \tilde{Q}(s_j, a'_j, (s, a)_{-j}) &= \sum_{i \in \mathbb{N}_j^m} (\tilde{Q}_i((s, a)_{\mathbb{N}_i^m}) \\ &\quad - \tilde{Q}_i(s_j, a'_j, (s, a)_{\mathbb{N}_i^m \setminus \{j\}})) \end{aligned}$$

Note that the right-hand side depends on the states and actions in the $2m$ -hop neighborhood of agent j . This is

because each $\tilde{Q}_i$ for $i \in \mathbb{N}_j^m$ depends on agent i 's m neighborhood, and hence giving an overall dependence on agent i 's $2m$ -hop neighborhood. If agent i were to receive the Q_j 's in its m -hop neighborhood and know the states and actions of everyone in its $2m$ -hop neighborhood, it would be able to compute what its own action is to follow the m -hop cooperative policy. There are also several algorithms that aim to solve optimization problems like in equation (9) in a decentralized manner, such as consensus and gossip algorithms (Shah, 2009), and the message-passing algorithms on graphs (Kschischang et al., 2002; Wainwright & Jordan, 2008). The class of m -hop cooperative policies encode various ways of cooperation between agents within their m -hop neighborhoods.

For a central planner who has access to the specifications of the entire system, the space complexity of storing and implementing these m -hop cooperative policies is only $\sum_i |\mathcal{S}_{\mathbb{N}_i^m}| |\mathcal{A}_{\mathbb{N}_i^m}|$, which is at best linear in the number of agents n . This is as opposed to the space complexity $|\mathcal{S}| |\mathcal{A}|$ of storing a general globally optimal policy that can scale, in the worst case, exponentially in n . Note that this is a natural generalization of the m -hop policy defined in the single-agent case in Theorem 1.

4.4. Existence result for two active agents

Our main result for the two-agent case shows the existence of an approximately-optimal m -hop cooperative policy with the error decaying exponentially with m .

For any subset $U \subseteq [n]$ and node $v \in [n]$, we define

$$\eta_v^U := \max_{\substack{s, a, \\ \bar{s}_U, \bar{a}_U}} d_{\text{TV}} \left(\mathbb{P}_v(\cdot \mid s, a), \mathbb{P}_v(\cdot \mid \bar{s}_U, s_{U^c}, \bar{a}_U, a_{U^c}) \right). \quad (10a)$$

Note that η_v^U measures the influence of state-action perturbations over the set U on the transition at node v . As a consequence of the one-hop assumption, we have $\eta_v^U = 0$ whenever $d_G(v, U) > 1$. Moreover, in terms of this notation, our original pairwise influence matrix $\mathbf{C}$ has entries $\mathbf{C}_{uv} = \eta_u^v$.

In addition to the original pairwise influence matrix $\mathbf{C}$, we define the n -dimensional mixed influence matrices $\mathbf{C}^{\text{both}}$, $\mathbf{C}^{\text{one}}$ and $\mathbf{C}^{\text{two}}$ with entries

$$\mathbf{C}_{uv}^{\text{both}} := \eta_u^{\{v, 1, 2\}}, \quad \mathbf{C}_{uv}^{\text{one}} := \eta_u^{\{v, 1\}}, \quad \text{and} \quad \mathbf{C}_{uv}^{\text{two}} := \eta_u^{\{v, 2\}}. \quad (10b)$$

In terms of the matrices $\{\mathbf{C}, \mathbf{C}^{\text{both}}, \mathbf{C}^{\text{one}}, \mathbf{C}^{\text{two}}\}$, we define the following two coefficients

$$\theta_{\text{club}} := \gamma \|\mathbf{C}^{\text{both}}\|_1 \|\mathbf{C}\|_1 \quad \text{and} \quad \theta_{\text{sep}} := \gamma \|\mathbf{C}^{\text{one}}\|_1 \|\mathbf{C}^{\text{two}}\|_1. \quad (10c)$$

For any subset $U \subseteq [n]$, define $c^U := (\eta_u^U)_{u \in [n]}$. Under the convention that duplicate coordinates are removed from the perturbation set, $c^{\{1, 2\}}$ is a column of $\mathbf{C}^{\text{both}}$, $c^{\{j\}}$ is the j th column of $\mathbf{C}$, and similarly $c^{\{1\}}$ and $c^{\{2, j\}}$ are columns of $\mathbf{C}^{\text{one}}$ and $\mathbf{C}^{\text{two}}$, respectively.

For a given player $i \in \{1, 2\}$, note that $\bar{i} := 3 - i$ corresponds to the other player. With this notation, we define the two graph distance functions

$$r_{\text{club}}(j) := \left(\left\lceil \frac{d_G(\{1, 2\}, \{j\})}{2} \right\rceil - 1 \right)_+ \quad \text{for } j \in [n], \quad \text{and} \quad (10d)$$

$$r_{\text{sep}}(i, j) := \left(\left\lceil \frac{d_G(\{i\}, \{\bar{i}, j\})}{2} \right\rceil - 1 \right)_+ \quad \text{for } i \in \{1, 2\}, j \in [n]. \quad (10e)$$

Theorem 2 (Existence of approximately-optimal local policy). *Given a two-agent γ -discounted graph-structured MDP, suppose that $\max\{\theta_{\text{club}}, \theta_{\text{sep}}\} < 1$. Then for every integer $m \geq 1$, there exists an m -hop local cooperative policy $\pi^{(m)}$ with the suboptimality gap*

$$\|Q^{\pi^{(m)}} - Q^*\|_{\infty} \leq \frac{2 \text{osc}(R)}{(1 - \gamma)^2} \sum_{i=1}^2 \sum_{j \notin \mathbb{N}_i^m} \min \left\{ \frac{(\theta_{\text{club}})^{r_{\text{club}}(j)}}{1 - \theta_{\text{club}}}, \frac{(\theta_{\text{sep}})^{r_{\text{sep}}(i, j)}}{1 - \theta_{\text{sep}}} \right\}, \quad (11)$$

See Section B for the proof.

The exponential decay in m in equation (11) can be seen more clearly with the following argument. We first note the inequality $\max\{d_G(\{1, 2\}, \{j\}), d_G(\{1\}, \{2, j\})\} \geq \lfloor m/2 \rfloor$, which holds for any $j \notin \mathbb{N}_1^m$ outside agent 1's m -hop neighborhood irrespective of the position of agent 2. Similarly, $\max\{d_G(\{1, 2\}, \{j\}), d_G(\{2\}, \{1, j\})\} \geq \lfloor m/2 \rfloor$ for any $j \notin \mathbb{N}_2^m$. This gives the bound

$$\|Q^{\pi^{(m)}} - Q^*\|_{\infty} \leq \frac{4n \text{osc}(R)}{(1 - \gamma)^2} \frac{(\max\{\theta_{\text{club}}, \theta_{\text{sep}}\})^{\lfloor \frac{m}{4} \rfloor - 1}}{1 - \max\{\theta_{\text{club}}, \theta_{\text{sep}}\}}. \quad (12)$$

Therefore, the bound decays exponentially for $m \geq 4$.

Comparing with Theorem 1, the error bound has a similar structure. For each agent i , there is a sum over the nodes outside the m -hop ball around it. For each agent ℓ outside the m -hop ball, the associated decay term characterizes the long-range effect of ignoring information outside the m -hop ball. The bound differs from Theorem 1 in the structure of this decay term. It trades off the effect of clubbing the two active agents together and decoupling them completely as a function of m . We also note the stronger condition on the influence matrix required here in terms of the ℓ_1 norm as opposed to the spectral radius. We clarify that it is possible

to write down a decay term that looks more similar to that in [Theorem 1](#) that only requires a bound on the spectral radius of the influence matrix. We chose to not do so for the sake of simplicity and to make the exponential decay in m clearer.

The main crux of the proof is in showing that the optimal Q -function Q^* can be approximated by a sum of Q -functions for each active agent that only depend on m -hop neighborhoods. The error terms in [Theorem 2](#) that arose due to ignoring information outside the $m/2$ -hop ball is precisely this approximation error in capturing Q^* .

Proof sketch: Here we provide some high-level comments on the proof of [Theorem 2](#); see [Section B](#) for the full details. The proof of this result is more involved than the single-agent guarantee in [Theorem 1](#). In the former case, we were able to directly analyze the dynamics of the projected Q -updates, and thereby bound the Bellman residual of the projected fixed point $Q^\dagger$.

In the multi-agent case, a direct generalization of the projected Q -iteration does not work. Naively fixing state-actions outside agent i 's m -hop neighborhood can lead to large projection errors, due to the possibility of agents lying outside the m -hop neighborhood that nonetheless have a strong effect on the optimal Q -function. We circumvent this challenge by instead constructing a surrogate function $\tilde{Q}^{(m)}$ that consists of a sum of projections onto m -sized neighborhoods. Our m -cooperative policy is then greedy with respect to this decomposition. However, the operations that define the surrogate $\tilde{Q}^{(m)}$ do *not* lead to a non-expansive operator, so that we cannot exploit a contraction analysis. Controlling the additional errors via different techniques leads to the extra factor of $(1 - \gamma)$ in the denominator of the bound in [Theorem 2](#).

Moreover, while the proof of [Theorem 2](#) is constructive, in the sense that we exhibit a particular m -surrogate $\tilde{Q}^{(m)}$, it is currently defined directly in terms of direct operations on the optimal function Q^* . It remains an open problem to determine an efficient algorithm for computing the surrogate $\tilde{Q}^{(m)}$, or a related quantity with similar guarantees.

4.5. Extension to k active agents

We now extend the two-agent result to the setting in which there are k active agents. Let $\mathcal{I} \subseteq [n]$ denote the active set, and write $k := |\mathcal{I}|$. Thus, $R_i = \mathbf{0}$ and $|\mathcal{A}_i| = 1$ for every $i \notin \mathcal{I}$.

For any subset $U \subseteq \mathcal{I}$, let $\bar{U} := \mathcal{I} \setminus U$, and define the mixed influence matrix $\mathbf{C}^U \in \mathbb{R}_+^{n \times n}$ by $(\mathbf{C}^U)_{u,v} := \eta_u^{U \cup \{v\}}$, $u, v \in [n]$, which tracks how much the transition kernel of state u is affected by changing the state-action of

every node in U and that of v simultaneously. In particular, $\mathbf{C}^\emptyset = \mathbf{C}$. Also define $\theta_U := \gamma \|\mathbf{C}^U\|_1 \|\mathbf{C}^{\bar{U}}\|_1$, and $\theta_\star := \max_{\emptyset \neq U \subseteq \mathcal{I}} \theta_U$.

For $i \in \mathcal{I}$ and $j \neq i$, define the collection of admissible cuts $\mathfrak{U}_{i,j} := \{U \subseteq \mathcal{I} \mid i \in U, j \notin U\}$ that assign nodes i and j into disjoint sets. For each such cut $U \in \mathfrak{U}_{i,j}$, define the minimum distance between the two disjoint sets and the associated correlation decay exponent as

$$d_U(j) := d_G(U, \bar{U} \cup \{j\}), \quad r_U(j) := \left(\left\lceil \frac{d_U(j)}{2} \right\rceil - 1 \right)_+.$$

With this notation, we can state our result that extends the result for two active agents to k active agents.

Corollary 1 (k active agents). *Consider a γ -discounted graph-structured MDP with active set $\mathcal{I}$ of cardinality k . Suppose that $\theta_\star < 1$. Then for every integer $m \geq 1$, there exists an m -hop cooperative policy $\pi^{(m)}$ such that*

$$\|Q^{\pi^{(m)}} - Q^*\|_\infty \leq \frac{2\text{osc}(R)}{(1 - \gamma)^2} \sum_{i \in \mathcal{I}} \sum_{j \notin \mathbb{N}_i^m} \min_{U \in \mathfrak{U}_{i,j}} \frac{\theta_U^{r_U(j)}}{1 - \theta_U}. \quad (13)$$

Moreover,

$$\|Q^{\pi^{(m)}} - Q^*\|_\infty \leq \frac{2\text{osc}(R)}{(1 - \gamma)^2} \frac{\theta_\star^{\lfloor m/2k \rfloor}}{1 - \theta_\star} \sum_{i \in \mathcal{I}} |[n] \setminus \mathbb{N}_i^m|. \quad (14)$$

Consequently for a given k , the suboptimality gap decays exponentially in m/k .

Comparing with [Theorem 2](#), the bound on the suboptimality gap here is a generalization of the bound there for k active agents. The proof of this result follows the construction and mixed-gap analysis used for the two-agent result. We elaborate on this in the appendix.

5. Numerical Results

In broad terms, the main take-aways from our theory are the existence of m -hop policies whose sub-optimality gap decays exponentially in m . In this section, we report the results of some simple numerical results that illustrate the form of these predictions for different graph topologies.

Binary infection model: Given a graph with n nodes, we considered a binary infection model with state space $\mathcal{S}_i = \{-1, +1\}$ at each node $i \in [n]$, with $s_i = +1$ corresponding to the infected state and $s_i = -1$ to the healthy state. Thus, the full state space $\mathcal{S} = \otimes_{i=1}^n \mathcal{S}_i$ has cardinality 2^n . Node 1 is associated with a binary action $a_1 \in \{0, 1\}$, where $a_1 = 1$ corresponds to a ‘‘treat’’ decision (e.g., a

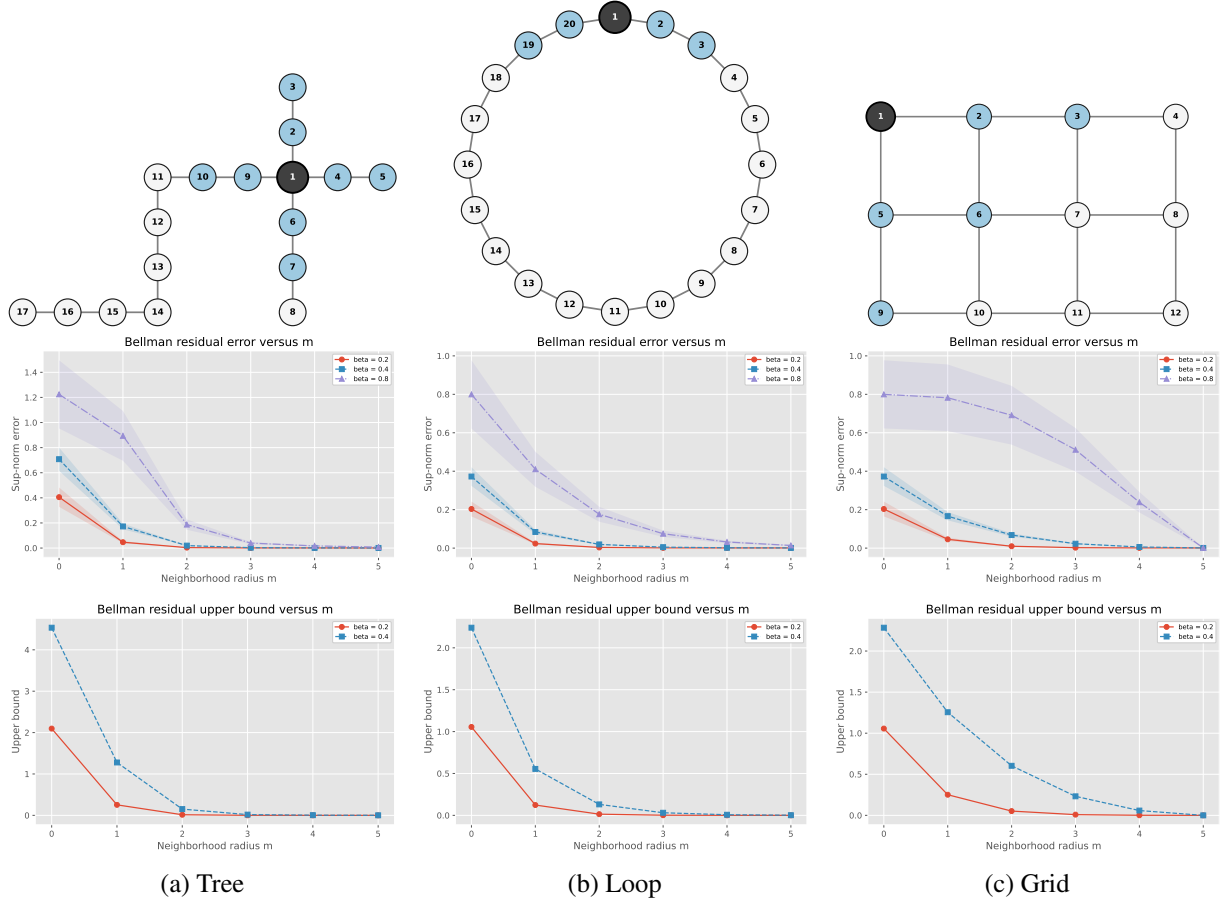


Figure 1. Behavior of m -hop local policies for three different graph types, and a simple binary infection model with the dynamics (15). Top row: illustrations of three different graphs, with active node 1 marked in black; the neighborhood set for hopsize $m = 2$ marked in blue; and remaining nodes in white. Middle row: Plots of the sup-norm Bellman residual error versus the m -hop policy size. Each curve corresponds to a different choice of connection strength β from the transition dynamics (15). Each plot shows the mean over ten trials with randomly constructed reward types, and shaded areas correspond to the standard errors over the reward randomness. Bottom row: Plots of the upper bound $U^{(m)}$ from equation (6b) versus the hopsize m . These plots are limited to the interaction pair $\beta \in \{0.2, 0.4\}$.

drug); all other nodes $j \neq 1$ have only $a_j = 0$. Letting $\sigma(t) = \frac{e^t}{1+e^t}$ denote the logistic function, the transition function $\mathbb{P}_i(s_i = +1 \mid s_{\mathbb{N}_i^1}, a_i)$ takes the form

$$\sigma\left(b + \rho s_i + \beta \sum_{j \in \mathbb{N}_i^1 \setminus \{i\}} s_j - \alpha a_i \mathbb{I}[s_i = 1]\right). \quad (15)$$

Here $b \in \mathbb{R}$ represents *base susceptibility* of individual i to the infection, with negative values biasing towards the uninfected state $s_i = -1$. The parameter $\rho \geq 0$ represents *individual infection persistence*, controlling how strongly the current state s_i , whether healthy or infected, propagates forward in time. The $\beta \geq 0$ represents the strength of *neighborhood infection propagation*, with larger values meaning greater probability of infection when many of node i 's neighbors are infected. Finally, the *treatment effect parameter* $\alpha > 0$ controls the strength of taking action $a_1 = 1$ to vaccinate. Note that α is irrelevant for any node $j \neq 1$, since in

this case we have $a_j = 0$ always.

In the simulations reported here, we implemented these dynamics with $b = -0.2$, $\rho = 0.6$, $\alpha = 1.0$, and the neighborhood strength β varying in the set $\{0.2, 0.4, 0.8\}$. In all cases, we instantiated agent rewards of the form

$$R_1(s_1, a_1) = \begin{matrix} s_1 = -1 & s_1 = +1 \end{matrix} \begin{bmatrix} a_1 = 0 & a_1 = 1 \\ 2.0 & 1.8 \\ -2.0 & -0.4 \end{bmatrix},$$

so that the treatment action $a_1 = 1$ has a strong positive effect (from -2.0 to -0.4) when in an infected state $s_1 = 1$, and a mild negative effect (from 2.0 to 1.8) when in a healthy state $s_1 = -1$. In order to generate ensembles, we perturbed the size of each effect size by a random uniform perturbation $\xi \in \text{Uni}[0, 0.5]$.

We performed simulations for three different graph topolo-

gies, as illustrated in the top row of Figure 1. For each graph, choice of β , reward perturbation, and neighborhood radius m , we computed the fixed point $Q_m^\dagger$ of the projected Bellman iteration $Q_{t+1}^{(m)} = \Pi^{(m)} \mathcal{B}^* Q_t^{(m)}$ up to accuracy $\epsilon = 0.001$. We then evaluated the theorem-relevant Bellman residual $\|\mathcal{B}^*(Q_m^\dagger) - Q_m^\dagger\|_\infty$ using the full graph transition kernel in the Bellman operator (3). For each graph and (β, m) -pair, we repeated this computation for 10 random perturbations of the reward function, and then computed the average sup-norm Bellman residual over these trials, along with 95% CIs over the 10 reward perturbations (to assess fluctuations with reward settings).

The panels in the middle row of Figure 1 give plots of these sup-norm residuals versus the neighborhood hopsizes m . Each plot corresponds to a different graph topology, with the three curves corresponding to the interaction strengths $\beta \in \{0.2, 0.4, 0.8\}$. Consistent with Theorem 1, each of these curves exhibit rapid decay as the hopsizes m is increased, with slower decay rates for larger values of β . The graph topology also plays a role, with the more densely connected grid graph exhibiting the slowest rates of convergence, especially for larger values of β . (For the grid graph, overly large values of β no longer lead to rapid decay, with the spectral radius condition being violated eventually.)

Finally, we numerically evaluated the upper bounds predicted by our theory; in particular, so as to compare directly to Bellman residuals, we compute the term $U^{(m)}$ defined in the refined upper bound (6b). The panels in the bottom row of Figure 1 give plots of this upper bound $U^{(m)}$ versus the hopsizes m . (We limited ourselves to plots for the interaction strengths $\beta \in \{0.2, 0.4\}$, since the spectral radius bound was not satisfied by all graphs for the setting $\beta = 0.8$.) In numerical terms, these bounds are conservative by a factor of 4 to 5 for the smaller hopsizes $m \in \{0, 1\}$, but exhibit the exponential decay guaranteed by our theory, and are qualitatively consistent with actual Bellman residuals in the middle row.

6. Discussion

In this paper, we studied the near-optimality of local policies for graph-structured MDPs, in particular those that depend only on the states in an m -hop neighborhood around a given agent. Our main results involve pairwise influence matrices that reflect the underlying graph topology, and capture the interaction strength between any two agents. We proved two main results that characterize the sub-optimality of m -local policies. For an influence matrix with suitably bounded spectral radius, we provide upper bounds on the sub-optimality that decay exponentially in the hopsizes m . In the multi-agent setting, we propose approximately-optimal policies that are defined by m -local functions, but require agents to communicate with their neighbors and coordinate

their actions. This is a rich class of policies that, to our knowledge, has not been studied before in the literature.

Our work opens multiple avenues for future study. In the single agent setting, we showed how to compute an approximately-optimal m -hop policy in space- and time-efficient manner. Results of such calculations were illustrated via a numerical study on an epidemic model in Section 5. In the more challenging multi-agent setting, we established the existence of an m -cooperative policy with bounds on sub-optimality, but it remains to give a computationally efficient procedure for computing such a greedy policy. Additionally, our analysis is limited to the regime where the number of active agents $k \leq m/2$. Finally, our analysis focused on the cooperative setting, in which all agents share the goal of maximizing a separable reward function over time. It would also be interesting to see if similar results hold for a competitive setting. Specifically, are there any notions of equilibria that can be approximated when each agent uses only local information?

Impact statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none that we feel must be specifically highlighted here.

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A. Proof of Theorem 1

We prove the stronger bound (6b) that involves the m -hop influence matrix $\mathbf{H}^{(m)}$ from equation (6a).

Our analysis exploits the following standard result (cf. Prop 2.2.1 in Bertsekas (2022)). Let π be a greedy policy induced by some function Q , and let Q^π be the Q -value function of the policy π . Then we have the uniform bound

$$\|Q^\pi - Q^*\|_\infty \leq \frac{2}{1-\gamma} \|\mathcal{B}^*(Q) - Q\|_\infty, \quad (16)$$

where $\mathcal{B}^*$ is the Bellman optimality operator (3). Thus, the Bellman residual $\mathcal{B}^*(Q) - Q$ can be used to control the sub-optimality of the greedy policy induced by Q .

Given the bound (16), we have reduced the problem to controlling the Bellman residual of a well-chosen Q -function. In particular, we consider the fixed point of a projected Bellman operator. For any hopsize $m = 0, 1, \dots$, we define the projection operator

$$(\Pi^{(m)}Q)(s, a_1) := Q(s_{\mathbb{N}_1^m}, \bar{s}_{n \setminus \mathbb{N}_1^m}, a_1),$$

which fixes all states outside the m -hop neighborhood to some fixed state $\bar{s}$. Observe that $\Pi^{(m)}$ is a non-expansive operator with respect to the ℓ_∞ -norm. As a result, the projected Bellman operator $\Pi^{(m)} \circ \mathcal{B}^*$ is γ -contractive in the ℓ_∞ -norm. The contraction mapping principle implies there is a unique $Q^\dagger$ satisfying the fixed point relation $\Pi^{(m)}(\mathcal{B}^*(Q^\dagger)) = Q^\dagger$, and moreover, that the iterates $Q_{t+1}^{(m)} = \Pi^{(m)}\mathcal{B}^*Q_t^{(m)}$ converge to $Q^\dagger$ from any initialization.

In order to complete the proof, it suffices to show that

$$\|\mathcal{B}^*(Q^\dagger) - Q^\dagger\|_\infty \leq \text{osc}(R_1) \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \left[\gamma^m \mathbf{C}^m (\mathbf{I} - \gamma \mathbf{C})^{-1} \right]_{(1,\ell)}. \quad (17a)$$

Our first step is to bound the differences $\Delta_{t+1}^{(m)}(s, a_1) := \mathcal{B}^*Q_t^{(m)}(s, a_1) - Q_{t+1}^{(m)}(s, a_1)$ along the iterates of the algorithm. In particular, we show that

$$\Delta_{t+1}^{(m)}(s, a_1) \leq \text{osc}(R_1) \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \left[\gamma^m \mathbf{C}^m (\mathbf{I} - \gamma \mathbf{C})^{-1} \right]_{(1,\ell)}, \quad (17b)$$

Note that the right-hand side of the bound (17b) is independent of the iteration number t , as well as the choice of (s, a_1) . Thus, the bound remains valid as we take $t \rightarrow \infty$, so that

$$[\mathcal{B}^*(Q^\dagger) - Q^\dagger](s, a) = \lim_{t \rightarrow \infty} \Delta_{t+1}^{(m)}(s, a_1) \leq \text{osc}(R_1) \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \left[\gamma^m \mathbf{C}^m (\mathbf{I} - \gamma \mathbf{C})^{-1} \right]_{(1,\ell)}.$$

Taking the supremum over (s, a_1) yields the ℓ_∞ -bound (17a).

Proof of the bound (17b): It remains to prove the intermediate claim (17b), which we do via two lemmas. For any Q -function Q and any node $\ell \in [n]$, define the local difference

$$\delta_\ell(Q) := \sup_{s_{-\ell}} \sup_{s_\ell, \tilde{s}_\ell} |Q(s_{-\ell}, s_\ell, a_1) - Q(s_{-\ell}, \tilde{s}_\ell, a_1)|. \quad (18)$$

This quantity tracks the effect of changing the state of node ℓ on Q . Our first lemma bounds the Bellman residual in terms of the matrix $\mathbf{H}^{(m)}$ and the local difference terms.

Lemma 1. *For each iteration $t = 1, 2, \dots$, the Bellman residual between successive iterates is upper bounded as*

$$\underbrace{\left| \left[\mathcal{B}^*Q_t^{(m)} - Q_{t+1}^{(m)} \right](s, a_1) \right|}_{\equiv |\Delta_{t+1}^{(m)}(s, a)|} \stackrel{(i)}{\leq} \sum_{\ell \in \mathbb{N}_1^m} \mathbf{H}_{1,\ell}^{(m)} \delta_\ell(Q_t^{(m)}) \stackrel{(ii)}{\leq} \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \delta_\ell(Q_t^{(m)}) \quad (19)$$

See [Section A.1](#) for the proof of this claim. Equality (ii) follows since $\mathbf{H}_{1,\ell}^{(m)} = 0$ for all nodes ℓ such $d_G(1, \ell) < m$, so we can restrict the sum to the shell $\mathbb{T}_1^m = \{\ell \in [n] \mid d_G(1, \ell) = m\}$.

Our next step is to bound the local difference terms.

Lemma 2 (Bounding $\delta_\ell(Q_t^{(m)})$). *For each iteration $t = 0, 1, 2, \dots$ and node $\ell \in [n]$, we have*

$$\delta_\ell(Q_t^{(m)}) \leq \text{osc}(R_1) \left\{ \mathbb{I}(\ell = 1) + \sum_{r=1}^t \gamma^r (\mathbf{C}^r)_{(1,\ell)} \right\}, \quad (20)$$

where $(\mathbf{C}^r)_{(1,\ell)}$ is the $(1, \ell)^{\text{th}}$ element of $\mathbf{C}^r$.

See [Section A.2](#) for the proof of this claim.

We now complete the proof by combining the pieces. Recall that $\mathbf{C}$ has the same zero pattern as the graph adjacency matrix. Consequently, for any $\ell \in \mathbb{T}_1^m$, we have $(\mathbf{C}^r)_{(1,\ell)} = 0$ all $r = 1, 2, \dots, m-1$, since the shortest path between node 1 and node ℓ has length m by definition of $\mathbb{T}_1^m$. Consequently, for any iteration $t \geq m$, the bound (20) implies that

$$\delta_\ell(Q_t^{(m)}) \leq \text{osc}(R_1) \sum_{r=m}^t \gamma^r (\mathbf{C}^r)_{(1,\ell)} \quad \text{for all } \ell \in \mathbb{T}_1^m.$$

Combining this upper bound with inequality (19) yields

$$\begin{aligned} \frac{\Delta_{t+1}^{(m)}(s, a)}{\text{osc}(R_1)} &\leq \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \sum_{r=m}^t \gamma^r (\mathbf{C}^r)_{(1,\ell)} = \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \left[\gamma^m \mathbf{C}^m \sum_{r=0}^{t-m} \gamma^r \mathbf{C}^r \right]_{(1,\ell)} \\ &\stackrel{(iii)}{\leq} \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \left[\gamma^m \mathbf{C}^m \sum_{r=0}^{\infty} \gamma^r \mathbf{C}^r \right]_{(1,\ell)} \\ &\stackrel{(iv)}{=} \sum_{\ell \in \mathbb{T}_1^m} \mathbf{H}_{1,\ell}^{(m)} \left[\gamma^m \mathbf{C}^m (\mathbf{I} - \gamma \mathbf{C})^{-1} \right]_{(1,\ell)}, \end{aligned}$$

where inequality (iii) follows since $\mathbf{C}$ has non-negative entries, and equality (iv) follows since the assumption $\gamma \rho(\mathbf{C}) < 1$ ensures that the geometric sum converges. This completes the proof of the bound (17b).

Recovering the weaker bound (5): In order to recover the weaker bound (5) stated in [Theorem 1](#), we need to show that $\mathbf{H}_{1,\ell}^{(m)} \leq \|\mathbf{C}\|_\infty$.

Consider the set $\mathcal{I}^m := [n] \setminus \mathbb{N}_1^m$ of nodes that lie outside agent 1's m -hop neighborhood. We have

$$\begin{aligned} \mathbf{H}_{1,\ell}^{(m)} &= \max_{\substack{s, \tilde{s} \\ a, \tilde{a}}} d_{\text{TV}} \left(\mathbb{P}_\ell(\cdot \mid s, a), \mathbb{P}_\ell(\cdot \mid (s, a)_{\mathbb{N}_1^m}, (\tilde{s}, \tilde{a})_{[n] \setminus \mathbb{N}_1^m}) \right) \\ &\stackrel{(i)}{\leq} \max_{\substack{s, \tilde{s} \\ a, \tilde{a}}} \sum_{j \in \mathcal{I}^m} d_{\text{TV}} \left(\mathbb{P}_\ell(\cdot \mid (s, a)_{\mathbb{N}_1^m}, (\tilde{s}, \tilde{a})_{v \in I_{-1}^m, v < j}, (s, a)_{v \in I_{-1}^m, v \geq j}), \mathbb{P}_\ell(\cdot \mid (s, a)_{\mathbb{N}_1^m}, (\tilde{s}, \tilde{a})_{v \in I_{-1}^m, v \leq j}, (s, a)_{v \in I_{-1}^m, v > j}) \right) \\ &\stackrel{(ii)}{\leq} \sum_{j \in \mathcal{I}^m} \mathbf{C}_{1,j} \\ &\leq \max_{k=1, \dots, [n]} \sum_{j=1}^n \mathbf{C}_{k,j} \stackrel{(iii)}{=} \|\mathbf{C}\|_\infty, \end{aligned}$$

where step (i) follows from the triangle inequality; step (ii) follows since each term in the sum depends only on the pairwise interaction between agents ℓ and v , which is controlled via $\mathbf{C}$; and step (iii) follows by the definition the induced ℓ_∞ -matrix operator norm.

A.1. Proof of Lemma 1

Using the update equation for $Q_{t+1}^{(m)}$, we can write

$$\begin{aligned}
 (\mathcal{B}^* Q_t^{(m)})(s, a_1) - Q_{t+1}^{(m)}(s, a_1) &= (\mathcal{B}^* Q_t^{(m)})(s, a_1) - (\Pi^{(m)} \mathcal{B}^* Q_t^{(m)})(s, a_1) \\
 &= (\mathcal{B}^* Q_t^{(m)})(s, a_1) - (\mathcal{B}^* Q_t^{(m)})(s_{\mathbb{N}_1^m}, \tilde{s}_{\mathbb{N}_1^{-m}}, a_1) \\
 &\stackrel{(i)}{=} \sum_{s'} (\mathbb{P}(s' | s, a_1) - \mathbb{P}(s' | s_{\mathbb{N}_1^m}, \tilde{s}_{\mathbb{N}_1^{-m}}, a_1)) \max_{\tilde{a}_1} Q_t^{(m)}(s', \tilde{a}_1) \\
 &\stackrel{(ii)}{=} \sum_{s'} (\mathbb{P}(s' | s, a_1) - \mathbb{P}(s' | s_{\mathbb{N}_1^m}, \tilde{s}_{\mathbb{N}_1^{-m}}, a_1)) \max_{\tilde{a}_1} Q_t^{(m)}(s'_{\mathbb{N}_1^m}, \tilde{a}_1) \\
 &\stackrel{(iii)}{=} \sum_{s'_{\mathbb{N}_1^m}} (\mathbb{P}(s'_{\mathbb{N}_1^m} | s_{\mathbb{N}_1^{m+1}}, a_1) - \mathbb{P}(s'_{\mathbb{N}_1^m} | s_{\mathbb{N}_1^m}, \tilde{s}_{\mathbb{N}_1^{m+1} \setminus \mathbb{N}_1^m}, a_1)) \max_{\tilde{a}_1} Q_t^{(m)}(s'_{\mathbb{N}_1^m}, \tilde{a}_1),
 \end{aligned} \tag{21}$$

where step (i) follows from the $R_1(s_1, a_1)$ term cancelling out as $m > 0$; step (ii) follows from the fact that each of the iterates $\{Q_t^{(m)}\}_{t \geq 0}$ only depends on the m -hop neighborhood of agent 1 as the projection operator fixes the states outside agent 1's m -hop neighborhood; and step (iii) follows from the decomposition (1c) of the transition kernel.

We now use the following lemma to upper bound the right-hand side and decouple the maximum. For each θ in some set Θ , let $f_\theta : \mathcal{Y} \rightarrow \mathbb{R}$ denote a real-valued function on the Cartesian product space $\mathcal{Y} := \otimes_{i=1}^k \mathcal{Y}_i$.

Lemma 3. *For any pair $\mathbb{Q}^{(1)}$ and $\mathbb{Q}^{(2)}$ of product measures over $\mathcal{Y} = \otimes_{i=1}^k \mathcal{Y}_i$, we have*

$$\sum_{y_1, \dots, y_k} (\mathbb{Q}^{(1)}(y_1, \dots, y_k) - \mathbb{Q}^{(2)}(y_1, \dots, y_k)) \max_{\theta} f_\theta(y_1, \dots, y_k) \leq \sum_i d_{TV}(\mathbb{Q}_i^{(1)}, \mathbb{Q}_i^{(2)}) \max_{y'_i, \tilde{y}_i, y_{-i}, \theta} |f_\theta(y'_i, y_{-i}) - f_\theta(\tilde{y}_i, y_{-i})|.$$

Variants of this lemma without the maximum operation underlie the classical Dobrushin analysis of mixing times (Dobrushin, 1970), and have been exploited in previous work on graph-structured MDPs (Qu et al., 2020). See Section A.3 for the proof.

Using the representation (21) of the Bellman residual in conjunction with Lemma 3, we have

$$\begin{aligned}
 (\mathcal{B}^* Q_t^{(m)})(s, a_1) - Q_{t+1}^{(m)}(s, a_1) &\leq \sum_{\ell \in \mathbb{N}_1^m} d_{TV}(\mathbb{P}_\ell(\cdot | s_{\mathbb{N}_1^{m+1}}, a_1), \mathbb{P}_\ell(\cdot | s_{\mathbb{N}_1^m}, \tilde{s}_{\mathbb{N}_1^{m+1} \setminus \mathbb{N}_1^m}, a_1)) \delta_\ell(Q_t^{(m)}) \\
 &\leq \sum_{\ell \in \mathbb{N}_1^m} \mathbf{H}_{1, \ell}^{(m)} \delta_\ell(Q_t^{(m)}),
 \end{aligned} \tag{22}$$

where the final inequality follows from the definition of the m -hop influence matrix $\mathbf{H}^{(m)}$.

A.2. Proof of Lemma 2

In order to prove this claim, we proceed via induction on the iteration number t . Beginning with $t = 0$, given our initialization $Q_0^{(m)} = \mathbf{0}$, we have $\delta_\ell(Q_0^{(m)}) = 0$, so that the bound (20) holds.

In the inductive step, we show that if the bound (20) holds for some t , then it also holds at time step $t + 1$. Using the update equation for $Q_{t+1}^{(m)}$, we can write

$$Q_{t+1}^{(m)}(s_\ell, s_{\mathbb{N}_1^m \setminus \{\ell\}}, a_1) - Q_{t+1}^{(m)}(s'_\ell, s_{\mathbb{N}_1^m \setminus \{\ell\}}, a_1) = T_1 + T_2$$

where $T_1 := \mathbb{I}(\ell = 1)(R_1(s_\ell, a_1) - R_1(s'_\ell, a_1))$, and

$$T_2 := \gamma \sum_{\tilde{s}_{\mathbb{N}_1^m}} (\mathbb{P}(\tilde{s}_{\mathbb{N}_1^m} | s_\ell, s_{\mathbb{N}_1^{m+1} \setminus \{\ell\}}, a_1) - \mathbb{P}(\tilde{s}_{\mathbb{N}_1^m} | s'_\ell, s_{\mathbb{N}_1^{m+1} \setminus \{\ell\}}, a_1)) \max_{\tilde{a}_1} Q_t^{(m)}(\tilde{s}_{\mathbb{N}_1^m}, \tilde{a}_1).$$

By definition of the oscillation, we have $|T_1| \leq \mathbb{I}(\ell = 1) \text{osc}(R_1)$.

Applying Lemma 3 to T_2 yields

$$|T_2| \leq \gamma \max_{s, a_1, s'_\ell} \sum_{j \in \mathbb{N}_1^m} d_{\text{TV}}(\mathbb{P}_j(\cdot | s_\ell, s_{\mathbb{N}_1^{m+1} \setminus \{\ell\}}, a_1), \mathbb{P}_j(\cdot | s'_\ell, s_{\mathbb{N}_1^{m+1} \setminus \{\ell\}}, a_1)) \delta_j(Q^t) \leq \gamma \sum_{j \in \mathbb{N}_1^m} \mathbf{C}_{j, \ell} \delta_j(Q_t^{(m)}),$$

We have upper bounded the Q^{t+1} difference for any choice of state-action, so we can take the maximum so as to bound $\delta_\ell(Q^{t+1})$. Doing so and combining our upper bounds on $|T_1|$ and $|T_2|$, we find that

$$\delta_\ell(Q_{t+1}^{(m)}) \leq \text{osc}(R_1) \mathbb{I}(\ell = 1) + \gamma \sum_{j \in \mathbb{N}_1^m} \mathbf{C}_{j, \ell} \delta_j(Q_t^{(m)}). \quad (23)$$

The differences $\delta_j(Q_t^{(m)})$ can be upper bounded using the induction hypothesis. Substituting these upper bounds into inequality (23), we find that

$$\begin{aligned} \delta_\ell(Q_{t+1}^{(m)}) &\leq \text{osc}(R_1) \mathbb{I}(\ell = 1) + \gamma \sum_{j \in \mathbb{N}_1^m} \mathbf{C}_{j, \ell} \text{osc}(R_1) (\mathbb{I}(j = 1) + \sum_{r=1}^t \gamma^r (\mathbf{C}^r)_{(1, j)}) \\ &\leq \text{osc}(R_1) \mathbb{I}(\ell = 1) + \gamma \text{osc}(R_1) \mathbf{C}_{1, \ell} + \gamma \text{osc}(R_1) \sum_{r=1}^t \gamma^r \sum_{j \in \mathbb{N}_1^m} \mathbf{C}_{j, \ell} (\mathbf{C}^r)_{(1, j)} \\ &= \text{osc}(R_1) \mathbb{I}(\ell = 1) + \gamma \text{osc}(R_1) \mathbf{C}_{1, \ell} + \text{osc}(R_1) \sum_{r=2}^{t+1} \gamma^r (\mathbf{C}^r)_{1, \ell} \\ &= \text{osc}(R_1) \left\{ \mathbb{I}(\ell = 1) + \sum_{r=1}^{t+1} \gamma^r (\mathbf{C}^r)_{1, \ell} \right\}, \end{aligned}$$

which completes the proof.

A.3. Proof of Lemma 3

Define the sequence of product measures

$$\mathbb{Q}^{(i)} := \left(\bigotimes_{j=1}^i \mathbb{Q}_j^{(1)} \right) \otimes \left(\bigotimes_{j=i+1}^k \mathbb{Q}_j^{(2)} \right), \quad \text{for } i = 0, 1, \dots, k.$$

As the index i increases from 0 to k , the measure $\mathbb{Q}^{(i)}$ is transformed from $\bigotimes_{j=1}^k \mathbb{Q}_j^{(2)}$ into $\bigotimes_{j=1}^k \mathbb{Q}_j^{(1)}$ on a product-by-product basis. Using this sequence, we have the telescoping relation

$$\bigotimes_{j=1}^k \mathbb{Q}_j^{(1)} - \bigotimes_{j=1}^k \mathbb{Q}_j^{(2)} = \sum_{i=1}^k (\mathbb{Q}^{(i)} - \mathbb{Q}^{(i-1)}),$$

from which it follows that

$$\sum_{y_1, \dots, y_k} (\mathbb{Q}^{(1)} - \mathbb{Q}^{(2)})(y_1, \dots, y_k) \max_{\theta} f_{\theta}(y_1, \dots, y_k) = \sum_{i=1}^k \sum_{y_1, \dots, y_k} (\mathbb{Q}^{(i)} - \mathbb{Q}^{(i-1)})(y_1, \dots, y_k) \max_{\theta} f_{\theta}(y_1, \dots, y_k).$$

For any given coordinate i , since $\mathbb{Q}^{(i)}$ and $\mathbb{Q}^{(i-1)}$ differ only in the i -th coordinate, we can write

$$(\mathbb{Q}^{(i)} - \mathbb{Q}^{(i-1)})(y) = (\mathbb{Q}_i^{(1)} - \mathbb{Q}_i^{(2)})(y_i) \prod_{j < i} \mathbb{Q}_j^{(1)}(y_j) \prod_{j > i} \mathbb{Q}_j^{(2)}(y_j).$$

Using this representation, we can write

$$\sum_{y_1, \dots, y_k} (\mathbb{Q}^{(i)} - \mathbb{Q}^{(i-1)})(y) \max_{\theta} f_{\theta}(y) = \sum_{y_{-i}} \prod_{j < i} \mathbb{Q}_j^{(1)}(y_j) \prod_{j > i} \mathbb{Q}_j^{(2)}(y_j) \sum_{y_i} (\mathbb{Q}_i^{(1)} - \mathbb{Q}_i^{(2)})(y_i) \max_{\theta} f_{\theta}(y_i, y_{-i}),$$

Since $\sum_{y_i} (\mathbb{Q}_i^{(1)} - \mathbb{Q}_i^{(2)})(y_i) = 0$, we have

$$\begin{aligned} \sum_{y_i} (\mathbb{Q}_i^{(1)} - \mathbb{Q}_i^{(2)})(y_i) \max_{\theta} f_{\theta}(y_i, y_{-i}) &= \sum_{y_i} (\mathbb{Q}_i^{(1)} - \mathbb{Q}_i^{(2)})(y_i) \left(\max_{\theta} f_{\theta}(y_i, y_{-i}) - \min_{\tilde{y}_i} \max_{\theta} f_{\theta}(\tilde{y}_i, y_{-i}) \right) \\ &\stackrel{(i)}{\leq} \sum_{y_i: \mathbb{Q}_i^{(1)} \geq \mathbb{Q}_i^{(2)}} (\mathbb{Q}_i^{(1)} - \mathbb{Q}_i^{(2)})(y_i) \left(\max_{\theta} f_{\theta}(y_i, y_{-i}) - \min_{\tilde{y}_i} \max_{\theta} f_{\theta}(\tilde{y}_i, y_{-i}) \right) \\ &\stackrel{(ii)}{\leq} d_{\text{TV}}(\mathbb{Q}_i^{(1)}, \mathbb{Q}_i^{(2)}) \max_{y_i, \tilde{y}_i, y_{-i}} \left| \max_{\theta} f_{\theta}(y_i, y_{-i}) - \max_{\theta} f_{\theta}(\tilde{y}_i, y_{-i}) \right|, \end{aligned}$$

where step (i) follows from dropping non-positive terms; and step (ii) follows from the definition $d_{\text{TV}}(\mathbb{P}, \mathbb{Q}) = \sum_u |\mathbb{P}(u) - \mathbb{Q}(u)|$ of the TV distance between measures $\mathbb{P}$ and $\mathbb{Q}$.

Taking the expectation over y_{-i} and upper bounding by the maximum yields

$$\sum_{y_1, \dots, y_k} (\mathbb{Q}^{(i)} - \mathbb{Q}^{(i-1)})(y) \max_{\theta} f_{\theta}(y) \leq d_{\text{TV}}(\mathbb{Q}_i^{(1)}, \mathbb{Q}_i^{(2)}) \max_{y'_i, \tilde{y}_i, y_{-i}, \theta} |f_{\theta}(y'_i, y_{-i}) - f_{\theta}(\tilde{y}_i, y_{-i})|.$$

Summing over $i = 1, \dots, k$ completes the proof.

B. Proof of Theorem 2

Here we prove our guarantee for the multi-agent setting.

B.1. Approximating Q^*

Our proof is based on exhibiting a surrogate Q -function $\tilde{Q}$ that is representable as a sum of two terms, one for each m -neighborhood around the two active agents. The m -cooperative maximization of $\tilde{Q}$ defines a greedy policy $\pi^{\dagger}$, and we bound its sub-optimality.

We begin by describing the construction of $\tilde{Q}$. Introduce the shorthand $z := (s, a)$ for state-action pairs, and introduce $\bar{z} := (\bar{s}, \bar{a})$ for some fixed state $\bar{s}$ and $\bar{a}$. Now consider the three intermediate state-action vectors

$$z^{(0)} := (\bar{z}_{\{1,2\}}, z_{-\{1,2\}}), \quad z^{(1)} := (z_1, \bar{z}_2, z_{-\{1,2\}}), \quad \text{and} \quad z^{(2)} := (z_1, z_2, z_{-\{1,2\}}) = z. \quad (24)$$

The choice of $z^{(0)}$ agrees with the current state-action pair outside the two active coordinates, but anchors the active coordinates to their reference values. Define the two active-coordinate increments $D_1^*(z) := Q^*(z^{(1)}) - Q^*(z^{(0)})$ and $D_2^*(z) := Q^*(z^{(2)}) - Q^*(z^{(1)})$. Then the telescoping identity gives

$$Q^*(z) = Q^*(z^{(0)}) + D_1^*(z) + D_2^*(z).$$

For each active agent $i \in \{1, 2\}$, let $\Pi_i^{(m)}$ denote projection onto the m -hop neighborhood of i , defined by $(\Pi_i^{(m)} f)(z) := f(z_{\mathbb{N}_i^m}, \bar{z}_{(\mathbb{N}_i^m)^c})$. We then define the projected increments

$$\tilde{D}_i(z) := (\Pi_i^{(m)} D_i^*)(z), \quad i \in \{1, 2\},$$

and the m -hop local surrogate

$$\tilde{Q}(z) := Q^*(z^{(0)}) + \tilde{D}_1(z) + \tilde{D}_2(z).$$

Hence $\tilde{Q}$ is a sum of three functions, where $Q^*(z^{(0)})$ has no action-dependence, and $\tilde{D}_i$ depends only on the state-action variables inside the m -hop neighborhood of active agent i . Therefore any policy that is greedy with respect to $\tilde{Q}$ is an m -hop cooperative policy, where each player only needs to coordinate with and use information from their m -hop neighborhood.

B.2. Analysis

Define the maximal action-oscillation of the function $F := Q^* - \tilde{Q}$ via

$$\|\text{osc}_a(Q^* - \tilde{Q})\|_\infty := \sup_s \sup_{a, a'} \{F(s, a) - F(s, a')\}. \quad (25)$$

The following result shows that controlling this quantity leads to bounds on the sub-optimality of the greedy policy $\pi^\dagger$.

Lemma 4 (From oscillation to performance guarantee). *Let $\pi^\dagger$ be a policy that is greedy with respect to $\tilde{Q}$, and suppose that $\|\text{osc}_a(Q^* - \tilde{Q})\|_\infty \leq \varepsilon$. Then the Q -function $Q^\dagger$ of $\pi^\dagger$ satisfies the bound*

$$\|Q^* - Q^\dagger\|_\infty \leq \frac{\varepsilon}{1 - \gamma}. \quad (26)$$

See [Section B.3.1](#) for the proof.

The remainder of the proof is devoted to bounding the maximal oscillation $\|\text{osc}_a(Q^* - \tilde{Q})\|_\infty$. In order to do so, it is convenient to introduce some intermediate quantities. For each active node $i \in \{1, 2\}$, we define the *coordinate advantage function*

$$\delta_i(Q^*)(s, a; \bar{s}_i, \bar{a}_i) := Q^*(s, a) - Q^*(\bar{s}_i, s_{-i}, \bar{a}_i, a_{-i}), \quad (27a)$$

which measures how changes in the state-action at node i affect the optimal Q -function Q^* . We then define the *decorrelation gap* between any two agents i and j by

$$\Delta_{i,j}(Q^*) := \max_{\substack{s, a, \\ \bar{s}, \bar{a}}} |\delta_i(Q^*)(s, a; \bar{s}_i, \bar{a}_i) - \delta_i(Q^*)((\bar{s}, \bar{a})_j, (s, a)_{-j}; \bar{s}_i, \bar{a}_i)|. \quad (27b)$$

This measures the effect that agent j 's state-action pair has on agent i 's coordinate advantage function. The following result shows how it bounds the maximum action oscillation.

Lemma 5 (Bounds on the action oscillation). *We have*

$$\|\text{osc}_a(Q^* - \tilde{Q})\|_\infty \leq \sum_{i=1}^2 \sum_{j \notin \mathbb{N}_i^m} \Delta_{i,j}(Q^*). \quad (28)$$

See [Section B.3.2](#) for the proof.

Our final step is to bound the gaps $\Delta_{i,j}(Q^*)$ in terms of the influence matrices, and in particular, in terms of the coefficients θ_{club} and θ_{sep} from equation (10c).

Lemma 6 (Bounds on the decorrelation gaps). *Under the stability condition $\max\{\theta_{\text{club}}, \theta_{\text{sep}}\} < 1$, we have*

$$\Delta_{1,j}(Q^*) \leq \frac{2 \text{osc}(R)}{1 - \gamma} \min \left\{ \frac{\theta_{\text{club}}^{r_{\text{club}}(j)}}{1 - \theta_{\text{club}}}, \frac{\theta_{\text{sep}}^{r_{\text{sep}}(1,j)}}{1 - \theta_{\text{sep}}} \right\} \quad (29)$$

with a similar statement for $\Delta_{2,j}$.

See [Section B.3.3](#) for the proof.

We now complete the proof of [Theorem 2](#). By [Lemma 6](#) and then [Lemma 5](#),

$$\|\text{osc}_a(Q^* - \tilde{Q})\|_\infty \leq \frac{2 \text{osc}(R)}{1 - \gamma} \sum_{i=1}^2 \sum_{j \notin \mathbb{N}_i^m} \min \left\{ \frac{(\theta_{\text{club}})^{r_{\text{club}}(j)}}{1 - \theta_{\text{club}}}, \frac{(\theta_{\text{sep}})^{r_{\text{sep}}(i,j)}}{1 - \theta_{\text{sep}}} \right\}.$$

Applying [Lemma 4](#) completes the proof.

B.3. Proofs of auxiliary lemmas

B.3.1. PROOF OF LEMMA 4

Let π^* denote an optimal policy, and let $\pi^\dagger$ denote the policy that is greedy with respect to $\tilde{Q}$. Fix a state s and introduce the shorthand $a^* = \pi^*(s)$ and $a^\dagger = \pi^\dagger(s)$. Since $a^\dagger$ is greedy for $\tilde{Q}$, we have $\tilde{Q}(s, a^*) \leq \tilde{Q}(s, a^\dagger)$. Therefore, in terms of the difference function $F = Q^* - \tilde{Q}$, we have

$$\begin{aligned} V^*(s) - Q^*(s, a^\dagger) &= Q^*(s, a^*) - Q^*(s, a^\dagger) = \{F(s, a^*) - F(s, a^\dagger)\} + \{\tilde{Q}(s, a^*) - \tilde{Q}(s, a^\dagger)\} \\ &\leq F(s, a^*) - F(s, a^\dagger) \\ &\leq \|\text{osc}_a(Q^* - \tilde{Q})\|_\infty \leq \varepsilon. \end{aligned}$$

Now use the Bellman equations under the action $a^\dagger = \pi^\dagger(s)$:

$$\begin{aligned} V^*(s) - V^\dagger(s) &= V^*(s) - Q^*(s, a^\dagger) + Q^*(s, a^\dagger) - Q^\dagger(s, a^\dagger) \\ &\leq \varepsilon + \gamma \mathbb{E}[V^*(s') - V^\dagger(s') \mid s, a^\dagger]. \end{aligned}$$

Taking suprema over s and rearranging gives $\|V^* - V^\dagger\|_\infty \leq \frac{\varepsilon}{1-\gamma}$. Finally, for every state-action pair (s, a) , we have $Q^*(s, a) - Q^\dagger(s, a) = \gamma \mathbb{E}[V^*(s') - V^\dagger(s') \mid s, a]$, from which the claimed bound $\|Q^* - Q^\dagger\|_\infty \leq \gamma \|V^* - V^\dagger\|_\infty \leq \frac{\varepsilon}{1-\gamma}$ follows.

B.3.2. PROOF OF LEMMA 5

Introduce the shorthand $E(z) := Q^*(z) - \tilde{Q}(z)$, and observe that $E(z) = \sum_{i=1}^2 \{D_i^*(z) - \tilde{D}_i(z)\}$. Fix a state s and two action profiles a, a' . Change the active actions from a' to a one active coordinate at a time. At the step corresponding to active coordinate i , the change in $D_i^* - \tilde{D}_i$ is a difference of the coordinate advantage of agent i evaluated under two configurations that agree on $\mathbb{N}_i^m$ and may differ only outside $\mathbb{N}_i^m$. Shell the outside coordinates one vertex at a time. By the definition of $\Delta_{i,j}$, the change caused by shelling vertex j is at most $\Delta_{i,j}(Q^*)$. Summing over $j \notin \mathbb{N}_i^m$ and then over $i = 1, 2$ yields the bound

$$|E(s, a) - E(s, a')| \leq \sum_{i=1}^2 \sum_{j \notin \mathbb{N}_i^m} \Delta_{i,j}(Q^*).$$

Taking the supremum over a, a' proves the claim.

B.3.3. PROOF OF LEMMA 6

To bound the decorrelation gaps, we first require some additional definitions. Let $z = (s, a)$ denote the full state-action vector. For two disjoint blocks A and B of state-action coordinates, by overloading the notation for the decorrelation gap (29), we define the mixed oscillation gap

$$\Delta_{A,B}(Q) := \max_{z_A, \bar{z}_A, z_B, \bar{z}_B, z_{[n] \setminus (A \cup B)}} |Q(z) - Q(\bar{z}_A, z_B, z_{[n] \setminus (A \cup B)}) - Q(z_A, \bar{z}_B, z_{[n] \setminus (A \cup B)}) + Q(\bar{z}_A, \bar{z}_B, z_{[n] \setminus (A \cup B)})|. \quad (30)$$

For future reference, we observe that

$$\Delta_{A,B}(Q) \leq \Delta_{A',B'}(Q), \text{ whenever } A \subseteq A' \text{ and } B \subseteq B'. \quad (31)$$

since the extra coordinates in $A' \setminus A$ and $B' \setminus B$ may be fixed to the same values in the barred and unbarred configurations. When the two blocks overlap, or contain duplicate coordinates, the corresponding mixed gap is interpreted after removing duplicate coordinates and is bounded by the trivial bound b below; the recursive argument only uses the nontrivial disjoint case.

For two agents i and j and action indices $p, q \in \{1, 2\}$, define the mixed coordinate effect function $\delta^{(s_i, a_p)}(s, a) := Q^*(s_i, a_p, s_{-i}, a_{\{1,2\} \setminus \{p\}}) - Q^*(\bar{s}_i, \bar{a}_p, s_{-i}, a_{\{1,2\} \setminus \{p\}})$, and the *mixed decorrelation gap*

$$\Delta^{(s_i, a_p), (s_j, a_q)} := \max_{\substack{s, a, \\ \bar{s}, \bar{a}}} \left| \delta^{(s_i, a_p)}(s, a) - \delta^{(s_i, a_p)}(\bar{s}_j, \bar{a}_q, s_{-j}, a_{\{1,2\} \setminus \{q\}}) \right|. \quad (32)$$

We use these quantities to bound decorrelation gaps $\Delta_{1,j}$ and $\Delta_{2,j}$; we focus on $\Delta_{1,j}$ for concreteness, since the argument for $\Delta_{2,j}$ is entirely analogous. We begin by observing that when $j \neq 2$,

$$\begin{aligned} \Delta_{1,j} &= \max_{\substack{s,a, \\ \bar{s}, \bar{a}}} |\delta_1(Q^*)(s, a; \bar{s}_1, \bar{a}_1) - \delta_1(Q^*)((\bar{s}, \bar{a})_j, (s, a)_{-j}; \bar{s}_1, \bar{a}_1)| \\ &\stackrel{(i)}{=} \max_{\substack{s,a, \\ \bar{s}, \bar{a}}} \left| \delta^{(s_1, a_1)}(s, a) - \delta^{(s_1, a_1)}((\bar{s}, \bar{a})_j, (s, a)_{-j}) \right| \\ &\stackrel{(ii)}{\leq} \min \left\{ \Delta^{(s_1, s_2, a_1, a_2), (s_j, a_j)}, \Delta^{(s_1, a_1), (s_2, a_2, s_j, a_j)} \right\}. \end{aligned} \quad (33)$$

Even though in this section we only assume that agents 1 and 2 have non-trivial actions, we write the above inequality for the general case where agent j also may have actions. The first term in the min in step (ii) follows from the monotonicity property (31): in the first-order oscillation in step (i), only the state-action of agent 1 is perturbed, and enlarging that block to also include the state-action of agent 2 can only increase the mixed gap. This gives $\Delta^{(s_1, s_2, a_1, a_2), (s_j, a_j)}$. The second term in the min follows in a similar way: as we only change the state-action of agent j in the second-order oscillation, by additionally perturbing the state-action of player 2 we get $\Delta^{(s_1, a_1), (s_2, a_2, s_j, a_j)}$.

Based on the bound (33), it suffices to bound the two types of mixed decorrelation gaps used there. Recall the definitions $\theta_{\text{club}} := \gamma \|\mathbf{C}^{\text{both}}\|_1 \|\mathbf{C}\|_1$ and $\theta_{\text{sep}} := \gamma \|\mathbf{C}^{\text{one}}\|_1 \|\mathbf{C}^{\text{two}}\|_1$, and the distances $r_{\text{club}}(j)$ and $r_{\text{sep}}(i, j)$.

Lemma 7 (Bounds on decorrelation gaps). *Under the stability condition $\max\{\theta_{\text{club}}, \theta_{\text{sep}}\} < 1$, we have*

$$\begin{aligned} \Delta^{(s_1, s_2, a_1, a_2), (s_j, a_j)} &\leq \frac{2\text{osc}(R)}{(1 - \theta_{\text{club}})(1 - \gamma)} \theta_{\text{club}}^{r_{\text{club}}(j)}, \\ \Delta^{(s_1, a_1), (s_2, a_2, s_j, a_j)} &\leq \frac{2\text{osc}(R)}{(1 - \theta_{\text{sep}})(1 - \gamma)} \theta_{\text{sep}}^{r_{\text{sep}}(1, j)}. \end{aligned}$$

If $j \notin \{1, 2\}$, then a_j belongs to a singleton action space and may be omitted from the notation.

See Section B.3.4 for the proof of this claim.

Combining Lemma 7 with (33) gives

$$\Delta_{1,j} \leq \min \left\{ \frac{2\text{osc}(R)}{(1 - \theta_{\text{club}})(1 - \gamma)} \theta_{\text{club}}^{r_{\text{club}}(j)}, \frac{2\text{osc}(R)}{(1 - \theta_{\text{sep}})(1 - \gamma)} \theta_{\text{sep}}^{r_{\text{sep}}(1, j)} \right\}.$$

B.3.4. PROOF OF LEMMA 7

For any two sets of vertices $A, B \subseteq V$, let $d_G(A, B) := \min_{i \in A, j \in B} d_G(i, j)$ be the minimum distance between any two vertices in those sets. Introduce the shorthand $b := \frac{2\text{osc}(R)}{1 - \gamma}$, and observe that since $\text{osc}(Q^*) \leq \text{osc}(R)/(1 - \gamma)$, every mixed second-order oscillation of Q^* is bounded by b . Define the n -dimensional matrices $\mathbf{F}$ and $\mathbf{G}$ with entries

$$F_{v,w} := \Delta^{(s_v, a_1, a_2), (s_w)}(Q^*), \quad \text{and} \quad G_{v,w} := \Delta^{(s_v, a_1), (s_w, a_2)}(Q^*),$$

and the two-hop boundary matrices $\mathbf{H}$ and $\mathbf{H}^{\text{both}}$ with entries

$$H_{v,w} := \mathbb{I}(d_G(\{v, 1, 2\}, \{w\}) \leq 2), \quad \text{and} \quad H_{v,w}^{\text{both}} := \mathbb{I}(d_G(\{v, 1\}, \{w, 2\}) \leq 2).$$

Lemma 8 (Recursive mixed-gap inequalities). *The mixed gaps satisfy the element-wise bounds*

$$\mathbf{F} \preceq b\mathbf{H} + \gamma \mathbf{C}^{\text{both}\top} \mathbf{F} \mathbf{C}, \quad \text{and} \quad \mathbf{G} \preceq b\mathbf{H}^{\text{both}} + \gamma \mathbf{C}^{\text{one}\top} \mathbf{G} \mathbf{C}^{\text{two}}. \quad (34)$$

Moreover, we have the series upper bounds

$$\mathbf{F} \preceq b \sum_{t=0}^{\infty} \gamma^t (\mathbf{C}^{\text{both}\top})^t \mathbf{H} \mathbf{C}^t, \quad \text{and} \quad \mathbf{G} \preceq b \sum_{t=0}^{\infty} \gamma^t (\mathbf{C}^{\text{one}\top})^t \mathbf{H}^{\text{both}} \mathbf{C}^{\text{two}t}. \quad (35)$$

Finally, for $d_{12,j} := d_G(\{1, 2\}, \{j\})$ and $d_{1,2,j} := d_G(\{1\}, \{2, j\})$, we have

$$\Delta^{(s_1, s_2, a_1, a_2), (s_j, a_j)} \leq b \mathbb{I}(d_{12,j} \leq 2) + \gamma (c^{\{1,2\}})^\top \mathbf{F} c^{\{j\}}, \quad \text{for } j \notin \{1, 2\}, \quad (36a)$$

$$\Delta^{(s_1, a_1), (s_2, a_2, s_j, a_j)} \leq b \mathbb{I}(d_{1,2,j} \leq 2) + \gamma (c^{\{1\}})^\top \mathbf{G} c^{\{2,j\}} \quad \text{for } j \neq 1. \quad (36b)$$

Using this auxiliary result, we now prove [Lemma 7](#). We split our proof into two parts, corresponding to the θ_{club} and θ_{sep} -based estimates.

Proof of θ_{club} -estimate: Define $d = d_{12,j} = d_G(\{1, 2\}, \{j\})$ and $r = (\lceil \frac{d}{2} \rceil - 1)_+$. If $j = 2$, then $r = 0$, and the desired clubbed bound follows from the trivial mixed-oscillation bound by b .

Otherwise, we can assume that $j \notin \{1, 2\}$. Applying [Lemma 8](#), combining the series bound (35) with the endpoint bound for the clubbed mixed gap (36a) yields the upper bound

$$\Delta^{(s_1, s_2, a_1, a_2), (s_j, a_j)} \leq b\mathbb{I}(d \leq 2) + b \sum_{t=0}^{\infty} \gamma^{t+1} (c^{\{1,2\}})^\top (\mathbf{C}^{\text{both}})^\top \mathbf{H} \mathbf{C}^t c^{\{j\}}.$$

Define the vectors $x_t = \mathbf{C}^{\text{both}^t} c^{\{1,2\}}$ and $y_t = \mathbf{C}^t c^{\{j\}}$. For nonnegative vectors x, y and a $\{0, 1\}$ -valued matrix H , $x^\top H y \leq \|x\|_1 \|y\|_1$. Since $c^{\{1,2\}}$ is a column of $\mathbf{C}^{\text{both}}$ and $c^{\{j\}}$ is a column of $\mathbf{C}$, we have

$$\gamma^{t+1} b x_t^\top \mathbf{H} y_t \leq b \gamma^{t+1} \|\mathbf{C}^{\text{both}}\|_1^{t+1} \|\mathbf{C}\|_1^{t+1} = b \theta_{\text{club}}^{t+1}.$$

It remains to identify which summands can be nonzero. For $U \subseteq V$, let $N^t(U)$ denote the set of vertices within graph distance at most t from U . By the one-hop support rule, we have $\text{supp}(x_t) \subseteq N^{t+1}(\{1, 2\})$ and $\text{supp}(y_t) \subseteq N^{t+1}(\{j\})$. Moreover, $H_{v,w} = 1$ implies $d_G(\{v, 1, 2\}, \{w\}) \leq 2$. Thus, if $x_t^\top \mathbf{H} y_t \neq 0$, then for some such pair (v, w) ,

$$d \leq d_G(\{1, 2\}, \{v\}) + d_G(\{v\}, \{w\}) + d_G(\{w\}, \{j\}) \leq (t+1) + 2 + (t+1).$$

Equivalently, with $\ell = t+1$, a summand can be nonzero only when $\ell \geq r$. The direct term $b\mathbb{I}(d \leq 2)$ occurs only when $r = 0$, and is then absorbed as the $\ell = 0$ term of the geometric series. Consequently,

$$\Delta^{(s_1, s_2, a_1, a_2), (s_j, a_j)} \leq b \sum_{\ell \geq r} \theta_{\text{club}}^\ell = \frac{b}{1 - \theta_{\text{club}}} \theta_{\text{club}}^r.$$

This proves the clubbed estimate.

Proof of θ_{sep} -estimate: For the separated estimate, set $d = d_{1,2,j} = d_G(\{1\}, \{2, j\})$ and $r = (\lceil \frac{d}{2} \rceil - 1)_+$. Applying [Lemma 8](#) to the separated mixed gap, combine the series bound (35) with the endpoint bound (36b). Define the vectors $\tilde{x}_t = \mathbf{C}^{\text{one}^t} c^{\{1\}}$ and $\tilde{y}_t = \mathbf{C}^{\text{two}^t} c^{\{2,j\}}$. The same norm bound gives

$$b \gamma^{t+1} \tilde{x}_t^\top \mathbf{H}^{\text{both}} \tilde{y}_t \leq b \theta_{\text{sep}}^{t+1}.$$

The support inclusions are now $\text{supp}(\tilde{x}_t) \subseteq N^{t+1}(\{1\})$ and $\text{supp}(\tilde{y}_t) \subseteq N^{t+1}(\{2, j\})$, and $H_{v,w}^{\text{both}} = 1$ implies $d_G(\{v, 1\}, \{w, 2\}) \leq 2$. Hence a nonzero summand again forces $d \leq 2t + 4$, or equivalently $\ell = t+1 \geq r$, while the direct term is present only when $r = 0$. Therefore, we have

$$\Delta^{(s_1, a_1), (s_2, a_2, s_j, a_j)} \leq b \sum_{\ell \geq r} \theta_{\text{sep}}^\ell = \frac{b}{1 - \theta_{\text{sep}}} \theta_{\text{sep}}^r,$$

which completes the proof of the separated estimate.

B.3.5. PROOF OF [LEMMA 8](#)

Proof of the recursions (34): We prove the recursion for $\mathbf{F}$; the proof for $\mathbf{G}$ is identical after replacing the clubbed influence matrices by the separated ones. Fix vertices v, w and consider the subsets $A = \{v, 1, 2\}$ and $B = \{w\}$.

If $d_G(A, B) \leq 2$, then the trivial bound on a mixed second-order oscillation gives

$$F_{v,w} \leq b \leq b H_{v,w} + \gamma (\mathbf{C}^{\text{both}} \mathbf{F} \mathbf{C})_{v,w},$$

since the second term is nonnegative.

Otherwise, we may assume that $d_G(A, B) > 2$. By the locality assumption on the reward, the immediate reward has zero mixed oscillation between the perturbations defining $F_{v,w}$. Introduce the shorthand $f(s, a) := \mathbb{E}_{s' \sim \mathbb{P}(\cdot | s, a)} [V^*(s') | s, a]$. By the Bellman equation, we have

$$F_{v,w} \leq \gamma \max |f(s_v, a_1, a_2, s_w, z) - f(\bar{s}_v, \bar{a}_1, \bar{a}_2, s_w, z) - f(s_v, a_1, a_2, \bar{s}_w, z) + f(\bar{s}_v, \bar{a}_1, \bar{a}_2, \bar{s}_w, z)|,$$

where the maximum is over two choices of (s_v, a_1, a_2) , two choices of s_w , and a common choice of all remaining state-action variables z .

Now fix two choices of (s_v, a_1, a_2) and two choices of s_w , keeping all other state-action coordinates fixed. Let $P_A^{(1)}$ and $P_A^{(2)}$ denote the one-step product transition laws induced by the two choices of (s_v, a_1, a_2) , and let $P_B^{(1)}$ and $P_B^{(2)}$ denote the analogous laws induced by the two choices of s_w . Since the transition kernel is a product kernel, the signed difference generated by changing (s_v, a_1, a_2) can be decomposed coordinatewise, and its total variation effect on coordinate u is bounded by $C_{u,v}^{\text{both}}$. Similarly, the effect of changing s_w on coordinate r is bounded by $C_{r,w}$. We will now require the following lemma to decouple the effect of the max operator (the proof for $\mathbf{G}$ will require a slightly different result which can be found in [Lemma 10](#)).

Lemma 9 (Clubbed two-action decoupling lemma).

$$\left| \sum_{x,y} (P_x^{(1)}(x) - P_x^{(2)}(x)) (P_y^{(1)}(y) - P_y^{(2)}(y)) \max_{a_1, a_2} Q(x, y, a_1, a_2) \right| \leq d_{\text{TV}}(P_x^{(1)}, P_x^{(2)}) d_{\text{TV}}(P_y^{(1)}, P_y^{(2)}) \Delta^{(x, a_1, a_2), (y)}(Q),$$

where

$$\Delta^{(x, a_1, a_2), (y)}(Q) := \max_{\substack{x, y, a_1, a_2, \\ \bar{x}, \bar{y}, \bar{a}_1, \bar{a}_2}} |\{Q(x, y, a_1, a_2) - Q(\bar{x}, y, \bar{a}_1, \bar{a}_2) - (Q(x, \bar{y}, a_1, a_2) - Q(\bar{x}, \bar{y}, \bar{a}_1, \bar{a}_2))\}|$$

The term $\Delta^{(x, a_1, a_2), (y)}(Q)$ here is a double oscillation term that measures how much the variable y affects the oscillation of Q with respect to x, a_1 and a_2 . The proof of this lemma can be found in [Section C.1](#).

Applying the clubbed two-action decoupling lemma to these signed coordinate transition differences gives

$$F_{v,w} \leq \gamma \sum_{u,r} C_{u,v}^{\text{both}} F_{u,r} C_{r,w}.$$

Together with the previous trivial-bound case, this proves that $F_{v,w} \leq bH_{v,w} + \gamma \sum_{u,r} C_{u,v}^{\text{both}} F_{u,r} C_{r,w}$, as claimed.

The proof of the recursion for $\mathbf{G}$ is the same. The first perturbation now has influence bounded by $C_{u,v}^{\text{one}}$ and the second by $C_{r,w}^{\text{two}}$, and the separated two-action decoupling inequality gives

$$\mathbf{G} \preceq b\mathbf{H}^{\text{both}} + \gamma[\mathbf{C}^{\text{one}}]^\top \mathbf{G} \mathbf{C}^{\text{two}}.$$

Proof of series bounds (35): Iterating the first recursion L times gives

$$\mathbf{F} \preceq b \sum_{t=0}^{L-1} \gamma^t ([\mathbf{C}^{\text{both}}]^\top)^t \mathbf{H} \mathbf{C}^t + \gamma^L ([\mathbf{C}^{\text{both}}]^\top)^L \mathbf{F} \mathbf{C}^L.$$

All matrices are nonnegative. Moreover $\mathbf{F}$ is bounded entrywise by b , and $\gamma \|\mathbf{C}^{\text{both}}\|_1 \|\mathbf{C}\|_1 = \theta_{\text{club}} < 1$. Hence the remainder term tends to zero entrywise as $L \rightarrow \infty$, proving

$$\mathbf{F} \preceq b \sum_{t=0}^{\infty} \gamma^t ([\mathbf{C}^{\text{both}}]^\top)^t \mathbf{H} \mathbf{C}^t.$$

The same argument, using $\gamma \|\mathbf{C}^{\text{one}}\|_1 \|\mathbf{C}^{\text{two}}\|_1 = \theta_{\text{sep}} < 1$, proves the series bound for $\mathbf{G}$.

Proof of endpoint inequalities: The endpoint inequalities (36a) and (36b) are the same one-step Bellman argument applied to the original mixed gaps. If the two endpoint blocks are within distance two, the immediate reward contributes at most b , producing the indicator terms. Otherwise the reward contribution is zero. The transition effect of perturbing $\{1, 2\}$ is bounded by $c^{\{1,2\}}$, while the effect of perturbing $\{j\}$ is bounded by $c^{\{j\}}$, giving

$$\Delta^{(s_1, s_2, a_1, a_2), (s_j, a_j)} \leq b\mathbb{I}(d_{12,j} \leq 2) + \gamma(c^{\{1,2\}})^\top \mathbf{F} c^{\{j\}}.$$

Likewise, perturbing $\{1\}$ and $\{2, j\}$ gives

$$\Delta^{(s_1, a_1), (s_2, a_2, s_j, a_j)} \leq b\mathbb{I}(d_{1,2j} \leq 2) + \gamma(c^{\{1\}})^\top \mathbf{G} c^{\{2,j\}}.$$

This proves all claims of the lemma.

B.4. Extension to k active agents

The proof for the k active agents case is a generalization of the two-agent case. Recall that in the proof for the two-agent case, we either clubbed the two active agents together and treated them as one, or treated them separately. Similarly, we will be partitioning the set of active agents into two disjoint sets here. There are three broad parts to the proof: we first construct a surrogate in terms of m -hop neighborhoods around the active agents that approximate the optimal Q^* . In the second part of the proof, we control the suboptimality error of this surrogate by partitioning the set of active agents into disjoint sets. In the final part of the proof, we solve for this suboptimality error by minimizing over all possible partitions.

Construction of the local surrogate. Fix an ordering of the active agents $\mathcal{I} = \{i_1, \dots, i_k\}$, and let $\bar{z} = (\bar{s}, \bar{a})$ be an arbitrary fixed state-action vector. For $r \in \{0, \dots, k\}$, define the set of r active agents $I_r := \{i_1, \dots, i_r\}$ and $z^{(r)} := (z_{I_r}, \bar{z}_{\mathcal{I} \setminus I_r}, z_{-\mathcal{I}})$ as the global state-action corresponding to fixing the state-actions of the active agents not in I_r . Thus, $z^{(0)} = (\bar{z}_{\mathcal{I}}, z_{-\mathcal{I}})$ and $z^{(k)} = z$. For $r \in \{1, \dots, k\}$, define the active-coordinate increment $D_r^*(z) := Q^*(z^{(r)}) - Q^*(z^{(r-1)})$ which tracks the effect of changing the state-action of just node i_r . By telescoping,

$$Q^*(z) = Q^*(z^{(0)}) + \sum_{r=1}^k D_r^*(z).$$

For $i \in \mathcal{I}$, let $\Pi_i^{(m)}$ denote projection onto the m -hop neighborhood of i : $(\Pi_i^{(m)} f)(z) := f(z_{\mathbb{N}_i^m}, \bar{z}_{(\mathbb{N}_i^m)^c})$. Define $\tilde{D}_r(z) := (\Pi_{i_r}^{(m)} D_r^*)(z)$, and $\tilde{Q}(z) := Q^*(z^{(0)}) + \sum_{r=1}^k \tilde{D}_r(z)$. The function $Q^*(z^{(0)})$ has no dependence on the active actions, while $\tilde{D}_r$ depends only on the state-action coordinates in $\mathbb{N}_{i_r}^m$. Therefore, any policy $\pi^\dagger$ that is greedy with respect to $\tilde{Q}$ is an m -hop cooperative policy.

Reduction to decorrelation gaps. Define $E(z) := Q^*(z) - \tilde{Q}(z)$. Using Lemma 4 gives

$$\|Q^{\pi^\dagger} - Q^*\|_\infty \leq \frac{1}{1-\gamma} \|\text{osc}_a(E)\|_\infty. \quad (37)$$

For each $i \in \mathcal{I}$, define the coordinate advantage $\delta_i(Q^*)(s, a; \bar{s}_i, \bar{a}_i) := Q^*(s, a) - Q^*(\bar{s}_i, s_{-i}, \bar{a}_i, a_{-i})$. Also define the corresponding decorrelation gap by

$$\Delta_{i,j}(Q^*) := \max_{\substack{s, a, \\ \bar{s}, \bar{a}}} \left| \delta_i(Q^*)(s, a; \bar{s}_i, \bar{a}_i) - \delta_i(Q^*)((\bar{s}, \bar{a})_j, (s, a)_{-j}; \bar{s}_i, \bar{a}_i) \right|.$$

The shelling argument in the proof of Lemma 5 holds here as well by summing over the active set $\mathcal{I}$:

$$\|\text{osc}_a(E)\|_\infty \leq \sum_{i \in \mathcal{I}} \sum_{j \notin \mathbb{N}_i^m} \Delta_{i,j}(Q^*). \quad (38)$$

The error associated with the increment of agent i is the difference between its coordinate advantage evaluated at two configurations that agree on $\mathbb{N}_i^m$. Shelling the coordinates outside $\mathbb{N}_i^m$ one at a time gives the inner sum in (38).

Reduction to an active-agent cut. For two disjoint blocks A and B of state-action coordinates, recall the definition of the mixed oscillation $\Delta^{A,B}(Q)$ (30). As in the two-agent proof, $\Delta^{A,B}(Q) \leq \Delta^{A',B'}(Q)$ whenever $A \subseteq A'$ and $B \subseteq B'$. Fix $i \in \mathcal{I}$, $j \neq i$, and $U \in \mathcal{U}_{i,j}$. Define $V := \mathcal{I} \setminus U$. Enlarging the perturbation at i to all active coordinates in U , and the perturbation at j to all active coordinates in V , gives $\Delta_{i,j}(Q^*) \leq \Gamma_{U,j}$, where

$$\Gamma_{U,j} := \Delta^{(s_U, a_U), (s_{V \cup \{j\}}, a_{V \cup \{j\}})}(Q^*).$$

Since the preceding inequality holds for every admissible cut,

$$\Delta_{i,j}(Q^*) \leq \min_{U \in \mathcal{U}_{i,j}} \Gamma_{U,j}. \quad (39)$$

Mixed-gap recursion for a fixed cut. Fix $U \in \mathcal{U}_{i,j}$ and let $V = \mathcal{I} \setminus U$. Recall the shorthand $b := \frac{2\text{osc}(R)}{1-\gamma}$ from the two-agent proof. Every mixed second-order oscillation of Q^* is uniformly bounded by b . Define the mixed-gap oscillation $F_{v,w}^U := \Delta^{(s_v, a_U), (s_w, a_V)}(Q^*)$, and the matrix $\mathbf{H}^U$ by $\mathbf{H}_{v,w}^U := \mathbb{I}[d_G(U \cup \{v\}, V \cup \{w\}) \leq 2]$. The same Bellman recursion and product-kernel decoupling argument used in the two-agent proof gives the elementwise inequality

$$F^U \preceq_{\text{e.w.}} b\mathbf{H}^U + \gamma(\mathbf{C}^U)^\top F^U \mathbf{C}^V. \quad (40)$$

As before, for each element (u, w) , if the two perturbation blocks $U \cup \{v\}$ and $V \cup \{w\}$ lie within distance two, the trivial mixed-oscillation bound gives the first term. Otherwise, the immediate reward cancels out in the mixed oscillation between the two blocks. The second term captures the effect of the mixed oscillation on the transition kernel. The effect on the kernel of node u by perturbing the state-action of the nodes in $U \cup \{v\}$ is bounded by $(\mathbf{C}^U)_{u,v} = \eta_u^{U \cup \{v\}}$, whereas the effect of perturbing the state-action of the nodes in $V \cup \{w\}$ on the transition kernel of node ℓ is bounded by $(\mathbf{C}^V)_{\ell,w} = \eta_\ell^{V \cup \{w\}}$. Finally, we use the two-action decoupling lemma (Lemma 10) with a_1 and a_2 replaced by the vectors a_U and a_V . This yields equation (40).

Iterating equation (40) and using $\theta_U < 1$ gives

$$F^U \preceq_{\text{e.w.}} b \sum_{t=0}^{\infty} \gamma^t ((\mathbf{C}^U)^\top)^t \mathbf{H}^U (\mathbf{C}^V)^t. \quad (41)$$

For any $S \subseteq [n]$, recall the notation $c^S := (\eta_u^S)_{u \in [n]}$, where each η_u^S is a vector. The endpoint Bellman argument gives

$$\Gamma_{U,j} \leq b\mathbb{I}[d_U(j) \leq 2] + \gamma(c^U)^\top F^U c^{V \cup \{j\}}. \quad (42)$$

Substituting equation (41) in the right-hand side yields

$$\Gamma_{U,j} \leq b\mathbb{I}[d_U(j) \leq 2] + b \sum_{t=0}^{\infty} \gamma^{t+1} ((\mathbf{C}^U)^t c^U)^\top \mathbf{H}^U ((\mathbf{C}^V)^t c^{V \cup \{j\}}). \quad (43)$$

Since U is nonempty, c^U is a column of $\mathbf{C}^U$. Moreover, $c^{V \cup \{j\}}$ is the j th column of $\mathbf{C}^V$. Consequently,

$$\|(\mathbf{C}^U)^t c^U\|_1 \leq \|\mathbf{C}^U\|_1^{t+1}, \quad \|(\mathbf{C}^V)^t c^{V \cup \{j\}}\|_1 \leq \|\mathbf{C}^V\|_1^{t+1}.$$

Since the entries of $\mathbf{H}^U$ are binary, the t th summand in equation (43) is at most

$$\gamma^{t+1} \|(\mathbf{C}^U)^t c^U\|_1 \|(\mathbf{C}^V)^t c^{V \cup \{j\}}\|_1 \leq \gamma^{t+1} \|\mathbf{C}^U\|_1^{t+1} \|\mathbf{C}^V\|_1^{t+1} = \gamma^{t+1} \theta_U^{t+1}. \quad (44)$$

As $\theta_U < 1$, the above bound implies that the right-hand side of equation (43) can be bounded via a geometric series. It remains to determine which of those summands can be nonzero, as that would establish the distance-dependent decay. For any $S \subseteq [n]$, define $\mathbb{N}^\ell(S) := \{v \in [n] \mid d_G(v, S) \leq \ell\}$ as the set of nodes at most ℓ hops away from S . As the transition kernel of each node only depends on its one-hop neighborhood,

$$\text{supp}((\mathbf{C}^U)^t c^U) \subseteq \mathbb{N}^{t+1}(U), \quad \text{supp}((\mathbf{C}^V)^t c^{V \cup \{j\}}) \subseteq \mathbb{N}^{t+1}(V \cup \{j\}).$$

This establishes that the nonzero entries of $(\mathbf{C}^U)^t \mathbf{c}^U$ are precisely those that are at most t hops away from U (and similarly for the other vector). If the t th summand in equation (43) is nonzero, then there exist v, w in the supports of $(\mathbf{C}^U)^t \mathbf{c}^U$ and $(\mathbf{C}^V)^t \mathbf{c}^{V \cup \{j\}}$ such that $\mathbf{H}_{v,w}^U = 1$. Therefore,

$$\begin{aligned} d_U(j) &= d_G(U, V \cup \{j\}) \\ &\leq d_G(U, v) + d_G(U \cup \{v\}, V \cup \{w\}) + d_G(w, V \cup \{j\}) \\ &\leq (t+1) + 2 + (t+1) \\ &= 2t+4. \end{aligned}$$

Writing $\ell = t+1$, a nonzero summand therefore requires

$$\ell \geq \left(\left\lceil \frac{d_U(j)}{2} \right\rceil - 1 \right)_+ = r_U(j).$$

Therefore, the terms in the right-hand side of (43) are zero till $t = r_U(j) - 1$. Combining this observation with equation (44) gives

$$\Gamma_{U,j} \leq b \sum_{\ell \geq r_U(j)} \theta_U^\ell = \frac{b}{1 - \theta_U} \theta_U^{r_U(j)}.$$

Recalling the definition of b and using equation (39) to bound the decorrelation gap, we obtain

$$\Delta_{i,j}(Q^*) \leq \frac{2\text{osc}(R)}{1 - \gamma} \min_{U \in \mathfrak{U}_{i,j}} \frac{\theta_U^{r_U(j)}}{1 - \theta_U}. \quad (45)$$

Combining equation (45) with the performance gap bounds in equations (38) and (37) proves equation (13).

Uniform exponential decay. Fix $i \in \mathcal{I}$ and $j \notin \mathbb{N}_i^m$, so that $d_G(i, j) > m$. Define $L := \lfloor \frac{m}{k} \rfloor$. Construct an auxiliary graph on the vertex set $\mathcal{I} \cup \{j\}$ by joining two vertices u and v whenever $d_G(u, v) \leq L$. The vertices i and j cannot belong to the same connected component. Otherwise, there would exist a simple auxiliary path from i to j containing at most k edges, and hence $d_G(i, j) \leq kL \leq m$, which contradicts $j \notin \mathbb{N}_i^m$.

Let U be the active vertices in the connected component containing i . Then $U \in \mathfrak{U}_{i,j}$. Moreover, every vertex in U has graph distance greater than L from every vertex in $\overline{U} \cup \{j\}$ (recall that $\overline{U} := \mathcal{I} \setminus U$). Since graph distances are integer-valued,

$$d_U(j) = d_G(U, \overline{U} \cup \{j\}) \geq L + 1 = \left\lfloor \frac{m}{k} \right\rfloor + 1.$$

It follows that

$$r_U(j) \geq \left\lceil \frac{\lfloor m/k \rfloor + 1}{2} \right\rceil - 1 = \left\lfloor \frac{m}{2k} \right\rfloor.$$

Since $\theta_U \leq \theta_\star < 1$,

$$\frac{\theta_U^{r_U(j)}}{1 - \theta_U} \leq \frac{\theta_\star^{\lfloor m/(2k) \rfloor}}{1 - \theta_\star}.$$

Substituting this inequality into equation (13) proves equation (14).

Remark 1. For $k = 2$, the cut $U = \mathcal{I}$ gives $\mathbf{C}^U = \mathbf{C}^{\text{both}}$ and $\mathbf{C}^{U^c} = \mathbf{C}$, recovering the clubbed estimate. The cut $U = \{i\}$ gives the matrices $\mathbf{C}^{\text{one}}$ and $\mathbf{C}^{\text{two}}$, recovering the separated estimate. Thus, Corollary 1 contains the two-agent argument as a special case.

C. Auxiliary lemmas

Here, we have the variant of the decoupling lemma where the effects of both actions are treated separately.

Lemma 10.

$$\left| \sum_{x,y} (P_x^{(1)}(x) - P_x^{(2)}(x)) (P_y^{(1)}(y) - P_y^{(2)}(y)) \max_{a_1, a_2} Q(x, y, a_1, a_2) \right| \leq d_{\text{TV}}(P_x^{(1)}, P_x^{(2)}) d_{\text{TV}}(P_y^{(1)}, P_y^{(2)}) \Delta^{(x, a_1), (y, a_2)}(Q),$$

where

$$\Delta^{(x, a_1), (y, a_2)}(Q) := \max_{\substack{x, y, a_1, a_2, \\ \bar{x}, \bar{y}, \bar{a}_1, \bar{a}_2}} | \{ Q(x, y, a_1, a_2) - Q(\bar{x}, y, \bar{a}_1, a_2) - (Q(x, \bar{y}, a_1, \bar{a}_2) - Q(\bar{x}, \bar{y}, \bar{a}_1, \bar{a}_2)) \} |$$

The proof of this claim can be found in [Section C.1](#). The version presented earlier in this work that clubbed the effects of both actions, [Lemma 9](#), follows directly from [Lemma 10](#) by considering the variable a_1 in [Lemma 10](#) to contain both the action variables a_1 and a_2 from [Lemma 9](#), and taking a_2 in [Lemma 10](#) to be from a singleton set.

C.1. Proof of [Lemma 10](#)

Let

$$f(x, y) := \max_{a_1, a_2} Q(x, y, a_1, a_2), \quad D := \Delta^{(x, a_1), (y, a_2)}(Q).$$

We first show that the mixed oscillation of f is at most D . Fix $x, \bar{x}, y, \bar{y}$. Let (a_1, a_2) be a maximizer in the definition of $f(x, y)$, and let $(\bar{a}_1, \bar{a}_2)$ be a maximizer in the definition of $f(\bar{x}, \bar{y})$. Then

$$\begin{aligned} f(x, y) - f(\bar{x}, y) - f(x, \bar{y}) + f(\bar{x}, \bar{y}) \\ \leq Q(x, y, a_1, a_2) - Q(\bar{x}, y, \bar{a}_1, a_2) - Q(x, \bar{y}, a_1, \bar{a}_2) + Q(\bar{x}, \bar{y}, \bar{a}_1, \bar{a}_2) \leq D. \end{aligned}$$

Interchanging x and $\bar{x}$ gives the reverse inequality, and therefore

$$|f(x, y) - f(\bar{x}, y) - f(x, \bar{y}) + f(\bar{x}, \bar{y})| \leq D. \quad (46)$$

Let

$$\mu_x = P_x^{(1)} - P_x^{(2)}, \quad \mu_y = P_y^{(1)} - P_y^{(2)}.$$

If either total variation distance is zero, the claim is immediate. Otherwise, write the Jordan decompositions as

$$\mu_x = \alpha(\mu_x^+ - \mu_x^-), \quad \mu_y = \beta(\mu_y^+ - \mu_y^-),$$

where $\alpha = d_{\text{TV}}(P_x^{(1)}, P_x^{(2)})$, $\beta = d_{\text{TV}}(P_y^{(1)}, P_y^{(2)})$, and $\mu_x^\pm, \mu_y^\pm$ are probability measures. Then

$$\sum_{x,y} \mu_x(x) \mu_y(y) f(x, y) = \alpha \beta \mathbb{E} [f(X^+, Y^+) - f(X^-, Y^+) - f(X^+, Y^-) + f(X^-, Y^-)],$$

where $X^\pm \sim \mu_x^\pm$ and $Y^\pm \sim \mu_y^\pm$ are independent. By equation (46), the absolute value of the expectation is at most D . Thus

$$\left| \sum_{x,y} \mu_x(x) \mu_y(y) f(x, y) \right| \leq d_{\text{TV}}(P_x^{(1)}, P_x^{(2)}) d_{\text{TV}}(P_y^{(1)}, P_y^{(2)}) D,$$

which proves the displayed bound.